

# PHRONESIS

AI-Powered Educational Decision Support for  
Teachers

Contents:

Contents

Contents:..... 0

Chapter 1: Introduction (975 words) ..... 3

**University of London CM3070 Final Project**  
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|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| 1.1 Introduction .....                                          | 3         |
| 1.2 Project Background.....                                     | 3         |
| 1.3 Problem Statement .....                                     | 3         |
| 1.4 Project Aims.....                                           | 4         |
| 1.5 Project Objectives .....                                    | 4         |
| 1.6 Project Motivation .....                                    | 4         |
| 1.7 Project Template .....                                      | 4         |
| 1.8 Intended Users.....                                         | 5         |
| 1.9 Project Scope .....                                         | 5         |
| 1.10 Project contribution.....                                  | 5         |
| 1.11 Report Structure .....                                     | 6         |
| Chapter 2 Literature Review (2498 Words).....                   | 6         |
| 2.1 Introduction .....                                          | 6         |
| 2.2 Teacher Workload and Data Burden .....                      | 6         |
| 2.3 Data-Informed Decision Making .....                         | 7         |
| 2.4 Existing Educational Information Systems .....              | 8         |
| 2.5 Educational Data Mining and Learning Analytics.....         | 12        |
| 2.6 Explainable AI .....                                        | 13        |
| 2.7 Educational Interventions .....                             | 13        |
| 2.8 Dashboard Design.....                                       | 14        |
| 2.9 Research Gap .....                                          | 14        |
| Chapter 3: System Design (1665 words) .....                     | 14        |
| 3.1 Introduction .....                                          | 14        |
| 3.2 Requirements Analysis.....                                  | 15        |
| 3.3 System Architecture.....                                    | 17        |
| <b>3.4 Data Design.....</b>                                     | <b>18</b> |
| 3.5 Explainability Engine Design .....                          | 21        |
| 3.6 Contextual Insight Design .....                             | 21        |
| 3.7 Recommendation Engine Design .....                          | 21        |
| 3.8 User Interface Design.....                                  | 22        |
| 3.9 Work Plan.....                                              | 30        |
| .....                                                           | 31        |
| 3.10 Evaluation Strategy .....                                  | 31        |
| Chapter 4: Implementation (1915 words).....                     | 32        |
| 4.1 Implementation Overview .....                               | 32        |
| 4.2 Development Environment and Final System Architecture ..... | 32        |

|                                                                    |    |
|--------------------------------------------------------------------|----|
| 4.3 Machine Learning Implementation .....                          | 33 |
| 4.4 Handling Incomplete and Missing Data .....                     | 36 |
| 4.5 Data Management and Validation .....                           | 37 |
| 4.6 Final Application Interface and Decision-Support Features..... | 39 |
| 4.7 Explanation and Recommendation Generation.....                 | 43 |
| 4.8 Accessibility and Usability Improvements .....                 | 43 |
| 4.9 Final Implementation Outcome .....                             | 46 |
| Chapter 5: Evaluation (2477 words).....                            | 46 |
| 5.1 Evaluation Overview and Approach.....                          | 46 |
| 5.2 Machine Learning Evaluation.....                               | 47 |
| 5.3 Functional and Integration Testing .....                       | 48 |
| 5.4 Edge-Case and Incomplete-Data Testing.....                     | 50 |
| 5.5 User Evaluation Method .....                                   | 51 |
| 5.6 User Evaluation Results.....                                   | 52 |
| 5.7 Evaluation Against Project Aims and User Requirements.....     | 53 |
| 5.8 Successes and Strengths .....                                  | 54 |
| 5.9 Limitations and Failures .....                                 | 54 |
| 5.10 Future Improvements and Extensions .....                      | 54 |
| 5.11 Evaluation Summary .....                                      | 55 |
| Chapter 6: Conclusion (601 words).....                             | 55 |
| 6.1 Project Outcome .....                                          | 55 |
| 6.2 What the Project Demonstrated.....                             | 55 |
| 6.3 Critical Reflection and Further Work .....                     | 55 |
| 6.4 Final Conclusion .....                                         | 56 |
| References: .....                                                  | 56 |
| APPENDIX A.....                                                    | 58 |
| Appendix A.1 Functional Requirements .....                         | 58 |
| Appendix A.2 Data Management Requirements .....                    | 59 |
| Appendix A.3 Future Requirements (Out of Scope).....               | 60 |
| Appendix A.4 Requirements Traceability Matrix .....                | 60 |
| APPENDIX B.....                                                    | 61 |
| Appendix B.1Teacher Survey Questions: .....                        | 61 |
| Appendix B.2 Survey Respondent Summary .....                       | 64 |
| Appendix B.3 Selected Survey results.....                          | 64 |

# Chapter 1: Introduction (975 words)

## 1.1 Introduction

Phronesis (*phrónēsis*), in Ancient Greek, refers to the type of wisdom or intelligence concerned with practical action. It is a web-based support and recommendation application which uses machine learning and data-processing techniques to identify patterns within educational datasets, predict academic risk, provide contextual insights and recommend intervention strategies.

This system aims to address a common issue found within schools, where substantial amounts of student data are collected and stored with the expectation of using it to support data-informed decision making. Existing school management information systems, such as SIMS, store and manage educational information and provide generalised student metrics effectively. However, they are not primarily designed to provide personalised explanations or intervention recommendations.

Rather than functioning as a traditional Management Information System (MIS), Phronesis acts as an intelligence layer that presents contextual indicators related to student performance and recommends possible interventions. This chapter introduces the final project, its motivation, aims, scope and intended users.

## 1.2 Project Background

Schools collect large amounts of educational data. As educational systems become increasingly datafied, schools are relying more on evidence-based decision making to support overall school improvement, with data-informed decision-making being recognised as important for "educational quality and management efficiency" (Jin et al., 2024).

Through advances in Educational Data Mining (EDM) and Machine Learning (ML), educational data can be analysed to identify patterns that may not be visible at first. Schools use Management Information Systems (MIS), such as SIMS and BromCom, increasing interest in technologies that transform educational data into meaningful insights.

## 1.3 Problem Statement

Teachers are expected to use data from attendance, behaviour, homework, and assessment records to inform decisions regarding student progress and wellbeing. While such information is readily available, teachers often find it challenging to interpret and translate it into meaningful actions to support learning. Studies on the educational data burden suggest that teachers frequently feel overwhelmed by "administrative and data-related responsibilities" (Morris et al., 2026).

Findings from the project survey and interviews highlighted similar challenges, with participants reporting difficulties interpreting student data, identifying patterns, and determining appropriate actions. Several respondents also described data collection as a "box-ticking" exercise, reflecting findings reported by IFF Research and UCL (2025).

Existing systems such as SIMS effectively manage educational information but still rely on teachers to interpret reports and identify patterns. Although Educational Data Mining demonstrates the potential to identify learning patterns and academic risk, much of the research focuses on prediction rather than supporting everyday teacher decision making. Consequently, there remains a need for systems that convert educational data into clear, explainable, and actionable recommendations.

## 1.4 Project Aims

The aim of this project is to develop an educational analytics application that supports teachers in making data-informed decisions by combining multiple educational data sources to identify students at academic risk and those who may require additional support. The platform will provide contextual insights and recommend possible intervention strategies to support, rather than replace, teacher judgement.

## 1.5 Project Objectives

The objectives for this project are:

- 1 To combine multiple educational data sources, including assessment, behaviour, homework, and attendance data, while incorporating behaviour and homework information as contextual indicators where available.
- 2 To create student profiles from the combined data.
- 3 To identify students who could be facing academic decline (at risk).
- 4 To provide contextual indicators to help teachers understand the factors associated to student performance.
- 5 To recommend appropriate intervention strategies based on available student information and risk status.
- 6 To develop an interactive dashboard to view student insights and tailored recommendations.
- 7 To prioritise explainable, text-based insights alongside concise visual summaries to support faster and more informed decision making.
- 8 To implement and evaluate a working teacher-facing decision-support application.
- 9 To handle incomplete student data transparently by avoiding predictions where required model information is unavailable.

## 1.6 Project Motivation

The motivation for this project stems from my experience as a high school mathematics teacher. I have seen instances where, based on the information available, a student was entered by the school, for a Foundation paper even though there were indications they could have been capable of attempting the Higher tier. Identifying these students often required manually reviewing of multiple sources of information alongside professional judgement.

I have observed that teachers balance data analysis alongside lesson planning, marking, and other professional responsibilities. This data is often difficult for teachers to translate into meaningful action that directly supports learning, a conclusion that was strengthened by the results of my surveys and interviews, where more than 60% of participants reported struggling with understanding student data and determining appropriate next steps.

This motivated the development of a system that helps teachers interpret student data and identify appropriate next steps.

## 1.7 Project Template

This project follows the CM3005 Data Science template 1.1 Project Idea 1: Data-Driven Personalised Educational Content Recommendation as it involves the collection, integration and analysis of educational data to identify patterns, predict academic risk and provide contextual insights and suggested interventions that support teacher decision making.

## 1.8 Intended Users

The intended users of the platform are primarily:

- Classroom teachers – The primary users, using Phronesis to review class/student information, identify academic risk and consider possible support. (e.g. arranging a parent-teacher meeting).
- Heads of Department - Secondary Users to monitor overall student progress across classes and identify wider trends through class-level summaries.
- Senior Leadership Teams- potential future users if the system were extended to provide wider school-level analytics.

## 1.9 Project Scope

To ensure the project remained achievable within the available timeframe, the scope of Phronesis has been limited to the features shown in Table 1.

| Included Features                                  | Excluded Features                                |
|----------------------------------------------------|--------------------------------------------------|
| Attendance analysis                                | Full MIS replacement                             |
| Assessment analysis                                | Timetable management                             |
| Student and class data management                  | Registration system                              |
| Subject-specific student profiles                  | Parent portal                                    |
| ML-based academic risk identification              | Live SIMS integration                            |
| At Risk / Not At Risk / Insufficient data handling | Multi-school deployment                          |
| Contextual student insights                        | Mobile application                               |
| Rule-based intervention recommendations            | Automatic decision making                        |
| Interactive dashboard and Class overview           | Production-grade database infrastructure         |
| Editable parent/carers email draft                 | Real-time school database connections            |
| Teacher-specific accounts/data separation          | Production deployment with sensitive school data |
| Accessibility settings and onboarding              |                                                  |

TABLE 1 PROJECT SCOPE

### 1.10 Project contribution

Phronesis contributes an educational decision-support system that combines educational data machine-learning risk classification, contextual indicators and suggested support strategies within one workflow. Unlike traditional MIS platforms, it prioritises actionable, text-based recommendations to support teacher decision making.

## 1.11 Report Structure

The remainder of this report is organised into three chapters. Chapter 2 presents the literature review. Chapter 3 describes the system design and evaluation strategy. Chapter 4 presents the Implementation, Chapter 5 evaluates technical and user performance, and Chapter 6 concludes the project and discusses further work.

## Chapter 2 Literature Review (2498 Words)

### 2.1 Introduction

This chapter will critically review the relevant literature for the development of Phronesis. The topics reviewed in this chapter were selected as they directly influence the design and functionality of the proposed system. It covers research on teacher workload, data-informed decision making, Educational Data Mining (EDM), Explainable Artificial Intelligence (XAI), existing school Management Information Systems (MIS), educational interventions, and dashboard design. This chapter also evaluates the strengths and limitations of the existing literature and technologies before identifying the research gap that Phronesis aims to address.

### 2.2 Teacher Workload and Data Burden

In education, teachers are increasingly being tasked with collecting and analysing student data to support decisions concerning students. Research suggests that the growing volume of data contributed to an increase in workload and challenges with interpreting the data effectively, despite the potential for data use to improve student progress. This section reviews the literature on teacher workload, the educational data burden, and implications for classroom practice.

#### 2.2.1 Increasing Data Collection

Compared to the past, schools now collect significantly more student data. Alongside assessment results, schools routinely record attendance, behaviour, homework, pastoral information and SEND information through digital technologies and Management Information Systems (MIS). This process of *datafication* has transformed schools into increasingly data-rich environments (King et al.).

Studies suggest that effective use of data can support "*school improvement and increased student achievement*" by using digital platforms that allows teachers to track individual student progress and identify trends over time (Jin et al., 2024; King et al., 2023). Beyond the classroom, schools' management also collect student data for accountability, and to also inform and evaluate the effectiveness of intervention strategies. However, the expansion of educational data has also increased the workload of teachers.

#### 2.2.2 Teacher Workload

As educational data has expanded, so have teachers' responsibilities. In addition to planning lessons, marking and classroom teaching, teachers are now expected to collect, manage, and analyse student data. Morris et al. (2026) describe this as the *educational data burden* and found that many teachers considered their workload to be "*unmanageable*", with work often extending into evenings, weekends, and holidays. The study also linked increasing workload with stress and reduced wellbeing. Similarly, IFF Research and IOE, UCL (2025) reported that administrative demands reduce the time available for lesson preparation and teaching. These findings

were reflected in the project survey, where one participant commented, *"Sometimes it can become tedious sifting through all students. If there was an automated way of identifying students of concern, it would be very helpful."*

A limitation of the existing research is that there is less agreement about whether increased data use improves teaching. While some researchers argue that data supports better decision making, others suggest the additional workload leaves teachers with less time to act on the information they collect.

### 2.2.3 Data Collection vs Data Use

Although pupil data has the potential to improve teaching, collecting information doesn't necessarily result in the most practical classroom action. Data can present patterns in student performance and support evidence-based decision making (King et al.; Jin et al., 2024). However, both studies place greater emphasis on the value of data than on how teachers can practically interpret it.

Researchers found that data collection is often viewed as a compliance or "box-ticking" activity rather than a process that directly supports learning (IFF Research & IOE, UCL, 2025). Similar views emerged from the project survey, where teachers reported recognising that students were struggling but remaining unsure of the underlying cause or the most appropriate response.

This suggests that the challenge facing schools is no longer collecting educational data but transforming it into information that teachers can readily understand and apply.

### 2.2.4 Critical Discussion

The body of the research indicates teachers have increased access to a large volume of data (King et al.; Morris et al., 2026). While Jin et al. (2024) mentions the advantages of data-informed-decision-making, Morris et al. (2026) and IFF Research and IOE, UCL (2025) highlight the additional workload involved in collecting and analysing data. This contrast suggests that while school data has a lot of potential to improve teaching, its advantages could be limited in the absence of tools to lessen the workload associated with information interpretation and decision-making.

## 2.3 Data-Informed Decision Making

Data-informed Decision Making (DIDM) is a key principle in modern education, encouraging teachers and schools to use evidence when making decisions about teaching and student progress. This section reviews how educational data supports decision making and the challenges associated with interpreting that information effectively.

### 2.3.1 Educational Data Use

Within a school, educational data is used to support decisions at various levels. Classroom teachers use assessment results, attendance, behaviour records, and homework completion to identify learning gaps and students who may need additional support. The information can then be used to adapt lesson planning, revisit topics that students have struggled with and provide additional explanations where needed.

Heads of Department use the same data differently. They can monitor progress across different classes, compare the performance of groups of students and review whether interventions have had a positive impact.

The data may also highlight areas where departments require additional support or changes to teaching approaches.

At a whole-school level, Senior Leadership Teams use educational data to monitor overall school performance, identify wider trends, and support strategic decisions such as resource allocation and school improvement planning (Pastore et al., 2026).

### *2.3.2 Challenges of Data Interpretation*

Educational data can support decision making; however, interpreting that information remains a significant challenge. Student performance is influenced by many different factors, meaning individual datasets rarely provide a complete picture. Instead, assessment results, attendance, behaviour, and homework records often need to be considered together before meaningful conclusions can be drawn.

This challenge was reflected in both the literature and the project findings. Several participants described difficulties identifying patterns in individual students' data and deciding what action to take afterwards. One teacher explained, "I can see there's a problem, but not always the cause," while another commented that there was "lots of data but not enough insight." These responses suggest that the difficulty lies not in accessing educational data, but in understanding what it means within the wider context of a student's learning.

A common limitation identified in both the literature and existing school systems is that they show teachers what the data is but rarely explain why a student may be struggling or what should happen next. Teachers are therefore responsible for interpreting the information, combining various sources of evidence, and deciding on the most appropriate support.

### *2.3.3 Critical Discussion*

Although educational data supports evidence-based decision making, its value depends on teachers being able to interpret and apply it effectively. Pastore et al. (2026) presents how school data can be used to support decision making at different levels of an organisation. However the literature and findings show that it is difficult to interpret multiple sources of student data. The widespread research focuses on collecting and presenting data, not on helping to use that data for interventions. This limitation provides a clear justification for decision-support systems that combine multiple sources of data, identify academic risk in order to support teachers in interpreting the results and deciding appropriate next steps.

## **2.4 Existing Educational Information Systems**

Existing Management Information Systems (MIS) play a vital role in storing and managing educational data. This section reviews three widely used systems to evaluate how effectively they support educational decision making.

### 2.4.1 SIMS

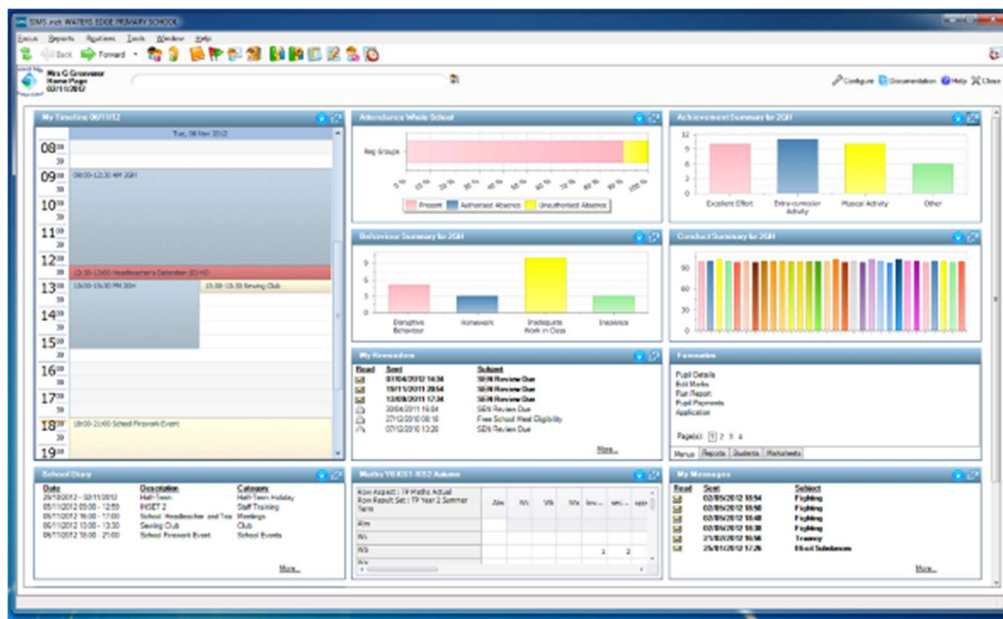


FIGURE 2.1: SIMS STUDENT OVERVIEW DASHBOARD (SOURCE: SIMS)

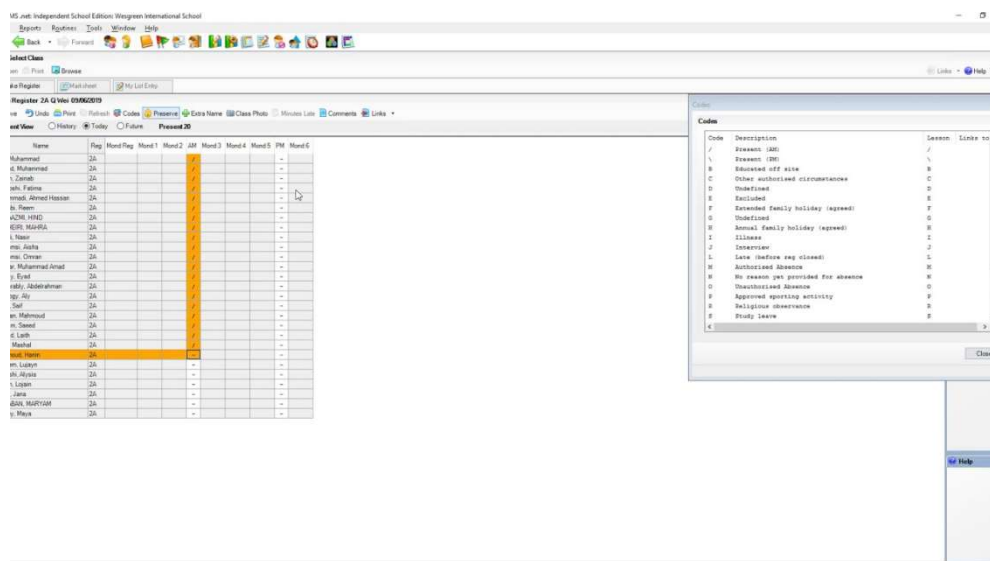
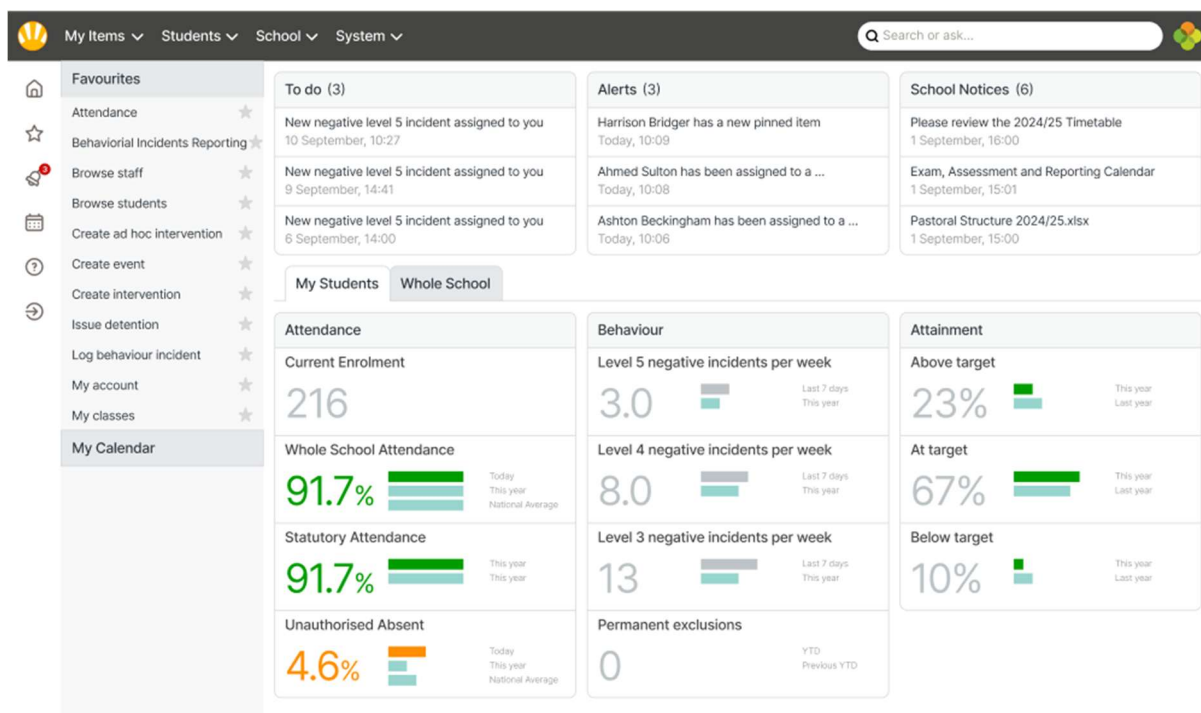


FIGURE 2.2: SIMS ATTENDANCE AND BEHAVIOUR PAGE (SOURCE: SIMS)

SIMS is one of the most widely used Management Information Systems (MIS) within UK schools. It provides a centralised platform for managing attendance, assessment, behaviour, and student records. As shown in Figure 2.1 and 2.2, SIMS presents copious amounts of student information through multiple menus and reporting tools. While this provides comprehensive access to educational data, users must still identify patterns and determine appropriate interventions themselves. Visscher et al. (2003) found that SIMS was "under-utilised considerably as a management tool", while user reviews on Capterra also report that the system can be difficult to navigate and requires substantial training.

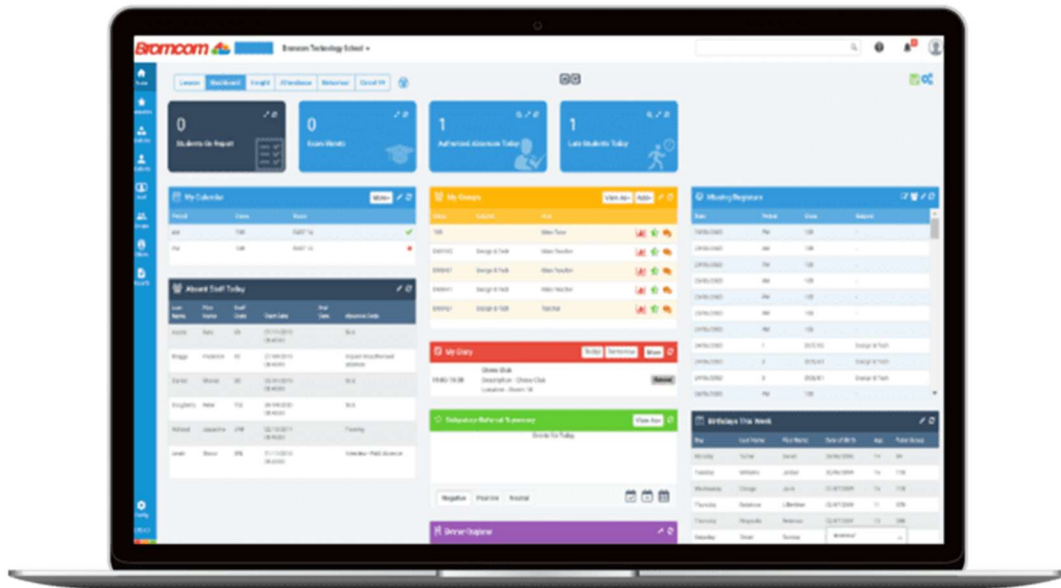
### 2.4.2 Arbor



**FIGURE 2.3: ARBOR DASHBOARD INTERFACE (SOURCE: ARBOR)**

Arbor is a cloud-based Management Information System designed with a more modern and user-friendly interface than SIMS. It manages attendance, behaviour, assessment, and communication through a single platform. As shown in Figure 2.3, information is displayed using dashboards and tiles, making key information easier to locate, reducing staff navigation time. Despite its improved usability, Arbor still focuses on presenting educational data rather than interpreting it, leaving teachers to identify at-risk students and determine appropriate actions.

### 2.4.3 BromCom



**FIGURE 2.4: BROMCOM ANALYTICS DASHBOARD (SOURCE: BROMCOM)**

Another popular cloud-based Management Information System is BromCom, which combines attendance, behaviour, assessment, reporting, and communication all into one platform. Compared with SIMS, it places greater emphasis on dashboard tables and analytics. The interface, as shown in Figure 2.4, provides quicker access to key performance information than traditional menu-based systems. User feedback also suggests that most prefer BromCom to older systems because of its dashboard layout and reporting features. However, its analytics remain highly descriptive, requiring teachers to interpret trends and decide on appropriate interventions themselves.

### 2.4.4 Comparative Analysis

| Feature                        | SIMS    | Bromcom  | Proposed AI Student Analytics Dashboard |
|--------------------------------|---------|----------|-----------------------------------------|
| Student Records                | ✓       | ✓        | ✓                                       |
| Attendance Tracking            | ✓       | ✓        | ✓                                       |
| Behaviour Tracking             | ✓       | ✓        | ✓                                       |
| Assessment Tracking            | ✓       | ✓        | ✓                                       |
| Reporting                      | ✓       | ✓        | ✓                                       |
| Student Analytics              | Limited | Moderate | Advanced AI Analytics                   |
| Dashboard Visualisations       | Basic   | Advanced | Interactive AI Dashboard                |
| Automated Trend Detection      | X       | X        | ✓                                       |
| Risk Prediction                | X       | Limited  | ✓                                       |
| Root Cause Analysis            | X       | X        | ✓                                       |
| AI Recommendations             | X       | X        | ✓                                       |
| Parent Email Generation        | X       | X        | ✓                                       |
| Intervention Suggestions       | X       | X        | ✓                                       |
| Personalised Student Summaries | X       | X        | ✓                                       |
| Predictive Analytics           | X       | Limited  | ✓                                       |

**TABLE 2: COMPARISON ON THE FEATURES OF SIMS, BROMCOM, ARBOR**

#### 2.4.5 Identified Gap

Although SIMS, Arbor and BromCom differ in their interface design and functionality, they share a common limitation. All three systems provide centralised data storage to assist school administration, however, despite these differences, the three platforms remain primarily focused on recording and presenting information rather than interpreting it. As shown in Table 2.1, The system provides limited support for personalised explanations. Therefore, teachers are still responsible for identifying patterns, understanding the causes of student performance, and deciding on appropriate interventions themselves.

## 2.5 Educational Data Mining and Learning Analytics

### 2.5.1 Educational Data Mining and Learning Analytics

Educational Data Mining (EDM) applies data mining, machine learning, and artificial intelligence techniques to discover "previously unknown information, relationships and patterns in large data repositories" (Kalita et al., 2025). These patterns can help identify learning habits, classify students into groups, detect unusual behaviour and support earlier educational interventions.

Learning Analytics builds on EDM by focusing on improving learning and teaching. Valtonen et al. (2025) found that teachers believed learning analytics helped them identify struggling students and target support more effectively once they had a better understanding of them. However, teachers also reported that many of the current dashboards focused on presenting data, with limited support for interpreting learning behaviours or deciding on appropriate actions.

### *2.5.2 Student Performance Prediction*

Student performance prediction is one of the most common applications of EDM. By analysing patterns with educational data, machine learning models can estimate a student's future performance and identify students at risk before it declines. Researchers found that previous attainment, demographics, learning behaviours and course characteristics all influence prediction accuracy (Duan et al., 2025). Combining multiple educational data sources can therefore improve prediction accuracy and support earlier intervention.

For teachers, these predictions enable earlier identification of students requiring additional support. However, Duan et al. (2025) also note that much of the existing research focuses on improving prediction accuracy rather than supporting classroom decision making. Consequently, predictions alone are of limited value unless teachers understand why a student has been identified as being at risk.

### *2.5.3 Limitations*

Although student performance prediction can identify students at risk, its reliability depends on the quality of the available data. Missing or inaccurate attendance, assessment or behaviour records may reduce prediction accuracy. These limitations highlight the need for transparent AI systems, discussed in the following section.

## **2.6 Explainable AI**

### *2.6.1 Explainable AI*

As Artificial Intelligence becomes increasingly integrated into education, there is growing recognition that its decisions should be understandable rather than simply accurate. Explainable Artificial Intelligence (XAI) aims to make AI-generated outputs transparent by showing how predictions and recommendations have been produced. Long et al. (2025) argue that many AI systems operate as "black boxes", making it difficult for users to understand or challenge their decisions. In educational settings, where decisions can influence student support and intervention, explainability is essential to ensure recommendations can be interpreted alongside professional judgement.

### *2.6.2 Trust and Human-Centred AI*

Teacher trust is a key factor influencing the successful adoption of AI. Shen et al. (2025) found that teachers were more willing to use AI when systems were transparent, easy to use and reduced workload, while uncertainty and privacy concerns discouraged adoption. Similarly, Aquilino et al. (2025) identify transparency, reliability and perceived usefulness as key factors affecting trust. Rather than replacing teachers, human-centred AI should support professional judgement by providing clear explanations while leaving final decisions to educators. These findings influenced Phronesis to present risk classifications alongside understandable contextual indicators and to retain teacher judgement as the final decision making authority.

## **2.7 Educational Interventions**

Educational interventions are most effective when they are targeted towards the specific needs of individual students. Attendance interventions have been shown to improve academic outcomes when implemented early and tailored according to the level of need (Melvin et al., 2025). Similarly, evidence from the Education Endowment Foundation (Higgins et al., 2012) highlights feedback, one-to-one tuition, and parental

engagement as among the most effective strategies for improving attainment. Cook et al. (2015) also demonstrated that targeted tutoring significantly improved mathematics achievement among disadvantaged secondary students.

The literature also suggests that early identification is essential, as interventions are generally more successful before patterns of underachievement become established. However, selecting the most appropriate intervention often relies on teacher judgement and experience. This highlights the potential value of decision-support systems that analyse multiple sources of student data and by suggesting possible interventions based on the available student information rather than relying on attainment alone.

## 2.8 Dashboard Design

Educational dashboards should help teachers understand information quickly and support effective decision making. Sedrakyan et al. (2019) argue that dashboards should do more than display student performance; they should identify at-risk students, provide meaningful feedback and guide intervention decisions. Poor dashboard design can increase cognitive load and information overload, making important patterns more difficult to identify. Consequently, dashboards should present only the most relevant information in a clear and accessible format. These principles support the development of Phronesis, which prioritises concise explanations and recommendations rather than large numbers of charts or reports.

## 2.9 Research Gap

The literature demonstrates that schools collect increasing amounts of educational data, and that Educational Data Mining and Artificial Intelligence can identify patterns and predict student performance. Existing Management Information Systems effectively store and display this information but provide limited support for interpreting it. There remains limited research on teacher-centred systems that combine predictive analytics with contextual student information and actionable decision support within a single workflow.

Phronesis aims to address this gap by integrating these components into a single application that identifies students at academic risk through ML, provide contextual factors contributing to those predictions and recommends evidence and rule-based support interventions. In doing so, it seeks to bridge the gap between educational data and practical classroom decision making.

# Chapter 3: System Design *(1665 words)*

## 3.1 Introduction

This chapter presents the design of Phronesis and explains how its features were developed to address the challenges identified in the literature review. Design choices are justified using findings from the literature, the project survey, teacher feedback, and interviews. The chapter covers the system requirements, architecture, data design, risk analysis, explainability and recommendation engines, user interface, and development plan.

## 3.2 Requirements Analysis

The system requirements stem from the literature review, teacher survey, interviews, and project objectives. Together, these sources identified the functionality required to support teacher decision making while ensuring the system remained practical, usable, and appropriate for everyday classroom use.

### 3.2.1 Functional Requirements

Functional requirements define the core features that Phronesis must provide to meet the needs identified through the literature review, teacher survey, and interviews. The requirements were prioritised using the MoSCoW method (Must, Should and Could) to distinguish essential functionality from desirable future enhancements. The complete requirements specification is provided in Appendix A.

| ID  | Requirement                  | Description                                                 | Source            | Priority |
|-----|------------------------------|-------------------------------------------------------------|-------------------|----------|
| FR1 | Dashboard Homepage           | Display key alerts and risk summaries.                      | Survey Q8/Q14     | Must     |
| FR2 | Student Profile              | Display combined student information.                       | Interviews        | Must     |
| FR3 | CSV Upload                   | Upload attendance, assessment, behaviour and homework data. | Survey            | Must     |
| FR4 | Risk Analysis                | Calculate student risk levels.                              | EDM Literature    | Must     |
| FR5 | AI Explanations              | Explain why students are identified as at risk.             | XAI Literature    | Must     |
| FR6 | Intervention Recommendations | Suggest evidence-based interventions.                       | Survey + EEF      | Must     |
| FR7 | Search & Filter              | Locate students efficiently.                                | Workload Research | Should   |
| FR8 | Parent Email Generator       | Generate draft parent emails.                               | Survey Q15        | Should   |

**TABLE 3.1 - FUNCTIONAL REQUIREMENTS**

The highest-priority requirements focus on supporting teacher decision making through student risk identification, explanation generation and intervention recommendations. These requirements directly reflect the needs identified during user research, where participants consistently requested tools that reduce interpretation time and provide clearer guidance for supporting students.

### 3.2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes of Phronesis rather than its functionality. These requirements ensure that the system is usable, dependable, secure, and appropriate for everyday classroom use. They were derived from the literature review, teacher feedback and recognised principles of educational dashboard design and Explainable Artificial Intelligence (XAI). However, in the implantation stage, Rule-based explanations were provided rather than XAI. The complete non-functional requirements specification is provided in Appendix A.

| ID    | Requirement        | Description                                                                   | Research Justification          | Priority |
|-------|--------------------|-------------------------------------------------------------------------------|---------------------------------|----------|
| NFR1  | Usability          | The interface should be simple and intuitive for teachers.                    | Teacher workload research       | Must     |
| NFR2  | Low Cognitive Load | Information should be presented clearly without overwhelming users.           | Dashboard design literature     | Must     |
| NFR3  | Explainability     | Every prediction and recommendation should include an explanation.            | XAI literature                  | Must     |
| NFR4  | Privacy            | Student data should be protected and anonymised during testing.               | Educational data ethics         | Must     |
| NFR5  | Accessibility      | The interface should use clear text and not rely solely on colour.            | Inclusive design principles     | Must     |
| NFR6  | Performance        | The dashboard should process class-sized datasets efficiently.                | Teacher usability requirements  | Should   |
| NFR7  | Maintainability    | The codebase should be modular to support future development.                 | Software engineering principles | Should   |
| NFR8  | Scalability        | The architecture should allow future expansion.                               | Future system growth            | Could    |
| NFR9  | Reliability        | Risk calculations should produce consistent outputs from the same input data. | Trust in AI research            | Must     |
| NFR10 | Data Integrity     | Uploaded data should remain accurate throughout processing.                   | Educational data management     | Must     |

**TABLE 3.2 – NON-FUNCTIONAL REQUIREMENTS**

### 3.2.3 User Stories

User stories were developed from the teacher interviews and survey findings to capture the primary interactions expected between users and the proposed system. They provide a user-centred view of the system requirements and guided the design of the application's core functionality.

| ID  | User Story                                                                                                                                           | Source                      |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| US1 | As a classroom teacher, I want to upload attendance, assessment, behaviour and homework data so that I can analyse student performance in one place. | Survey Q4; Interviews       |
| US2 | As a classroom teacher, I want the system to automatically identify students at academic risk so that I can prioritise support.                      | Survey Q11; Interview 3     |
| US3 | As a classroom teacher, I want explanations for each risk prediction so that I understand why a student has been identified.                         | Survey Q8, Q14; Interview 4 |
| US4 | As a classroom teacher, I want evidence-based intervention recommendations so that I know what action to take next.                                  | Survey Q8; EEF Literature   |
| US5 | As a Head of Department, I want to monitor student performance across classes so that I can identify wider trends and evaluate interventions.        | Interviews                  |
| US6 | As a Senior Leader, I want an overview of student performance so that I can support school improvement and allocate resources effectively.           | Literature; Interviews      |

**TABLE 3.3- USER STORIES**

### 3.2.4 Requirements Traceability Matrix

A Requirements Traceability Matrix (RTM) was produced to ensure that all high-priority system requirements could be traced through the design process. This helps verify that each requirement is addressed by a corresponding system component or design activity.

| Requirement ID | Requirement                  | Design Section            | Prototype Feature           |
|----------------|------------------------------|---------------------------|-----------------------------|
| FR1            | Dashboard Homepage           | 3.8 User Interface Design | Dashboard overview          |
| FR3            | Student Profile              | 3.8 User Interface Design | Student profile page        |
| FR9            | Risk Score                   | 3.5 Risk Analysis Engine  | Risk calculation engine     |
| FR10           | Explainability Engine        | 3.6 Explainability Engine | Plain-language explanations |
| FR13           | Intervention Recommendations | 3.7 Recommendation Engine | Suggested interventions     |
| FR19           | CSV Data Upload              | 3.4 Data Design           | CSV upload process          |
| FR25           | Data Validation              | 3.4 Data Design           | Input validation            |
| NFR3           | Explainability               | 3.6 Explainability Engine | Explainable outputs         |

TABLE 3.4: REQUIREMENTS TRACEABILITY MATRIX

### 3.3 System Architecture

Phronesis' will initially be a web-based analytics system that will allow teachers to upload student data, process it through analytic engines, and then view results on an interactive page. The system follows a modular structure so upload, risk analysis, explanations, and recommendations can be developed and tested separately.

#### 3.3.1 Initial Prototype: High-Level Architecture

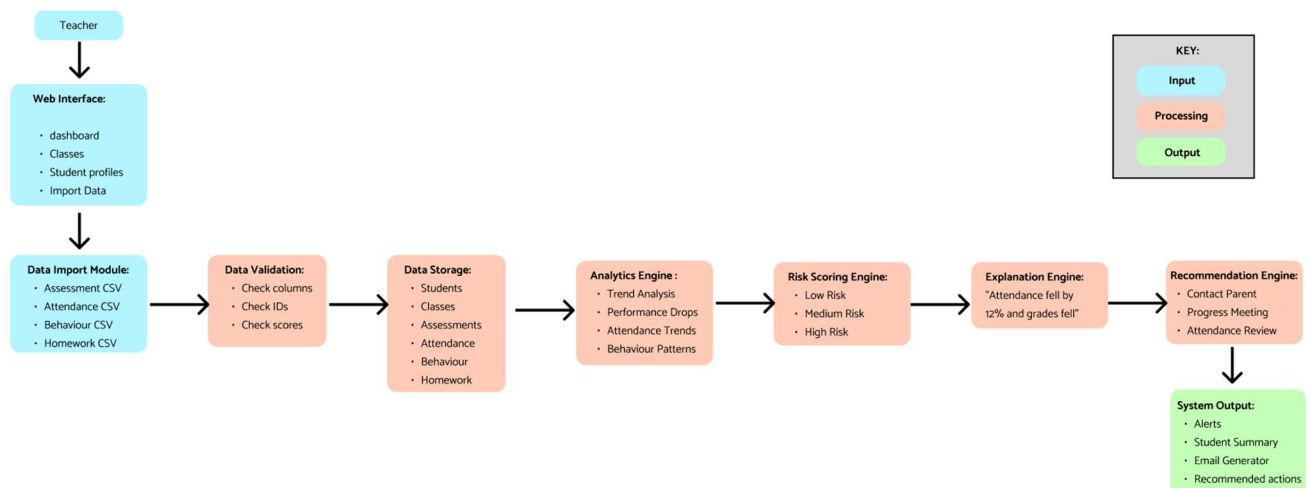


FIGURE 3.1: SYSTEM ARCHITECTURE DIAGRAM

As shown in Figure 3.1, Phronesis follows a modular architecture consisting of three stages: input, processing, and output. Teachers upload attendance, assessment, behaviour, and homework data through the web interface, where it is validated before being analysed. The processed data is passed through the risk scoring, explanation, and recommendation engines to generate student summaries, alerts, and suggested interventions.

### 3.3.1 Final Prototype: High-Level Architecture

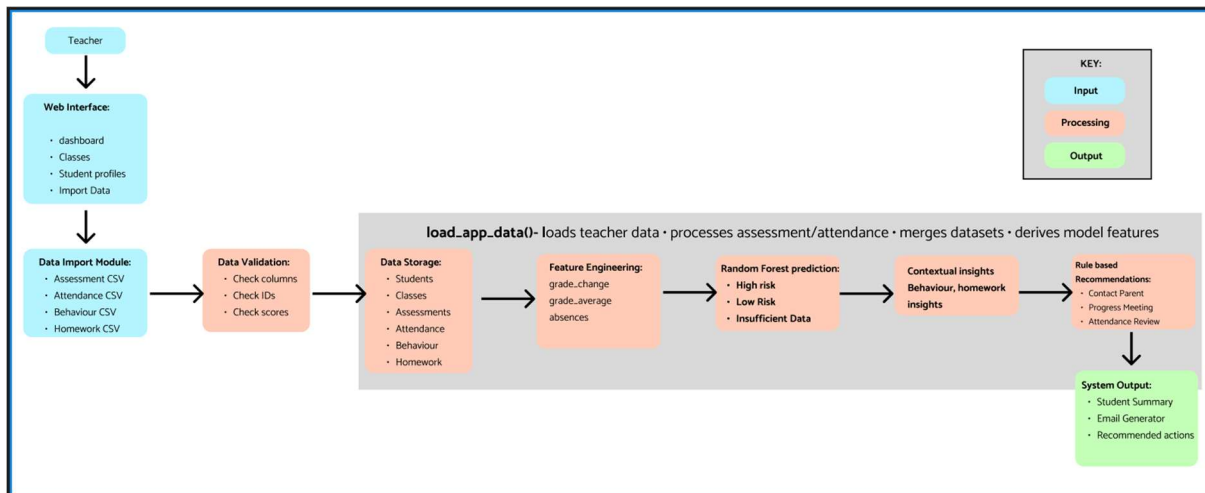


FIGURE 3.2.1 FINAL PROTOTYPE SYSTEM ARCHITECTURE

Figure 3.2.1 shows the final Phronesis architecture. Teacher-entered data is validated and stored before `load_app_data()` acts as the central processing pipeline, loading the current teacher's records, merging datasets, engineering model features and applying the Random Forest prediction. The resulting risk status is then combined with contextual insights and rule-based recommendations before being presented through the dashboard and student views. This modular design improves maintainability by allowing individual components to be updated or extended independently without affecting the overall system.

### 3.3.2 Front-End Design

The front end of Phronesis was developed using Streamlit to provide an interactive web interface that teachers can access through a standard web browser without the need for installation. The interface was designed to present educational data in a simple and organised manner, reducing navigation across multiple screens. It provides access to the dashboard, student profiles, and data upload pages. The detailed interface design is presented in Section 3.8.

### 3.3.3 Back-End Design

The back end of Phronesis was developed in Python using the Pandas library to process educational data. After the uploaded CSV files are validated, the system combines attendance, assessment, behaviour, and homework records into a single dataset. This dataset is then used to calculate student risk levels, generate explanations and produce intervention recommendations before the results are displayed through the web interface.

## 3.4 Data Design

The data model was designed to combine multiple sources of educational data into a unified student profile. Rather than using a relational database, the prototype stores information in separate CSV files representing subjects, classes, students, assessments and attendance, with behaviour and homework retained as contextual

fields where available. This reflects the way many schools already manage student information while simplifying prototype development.

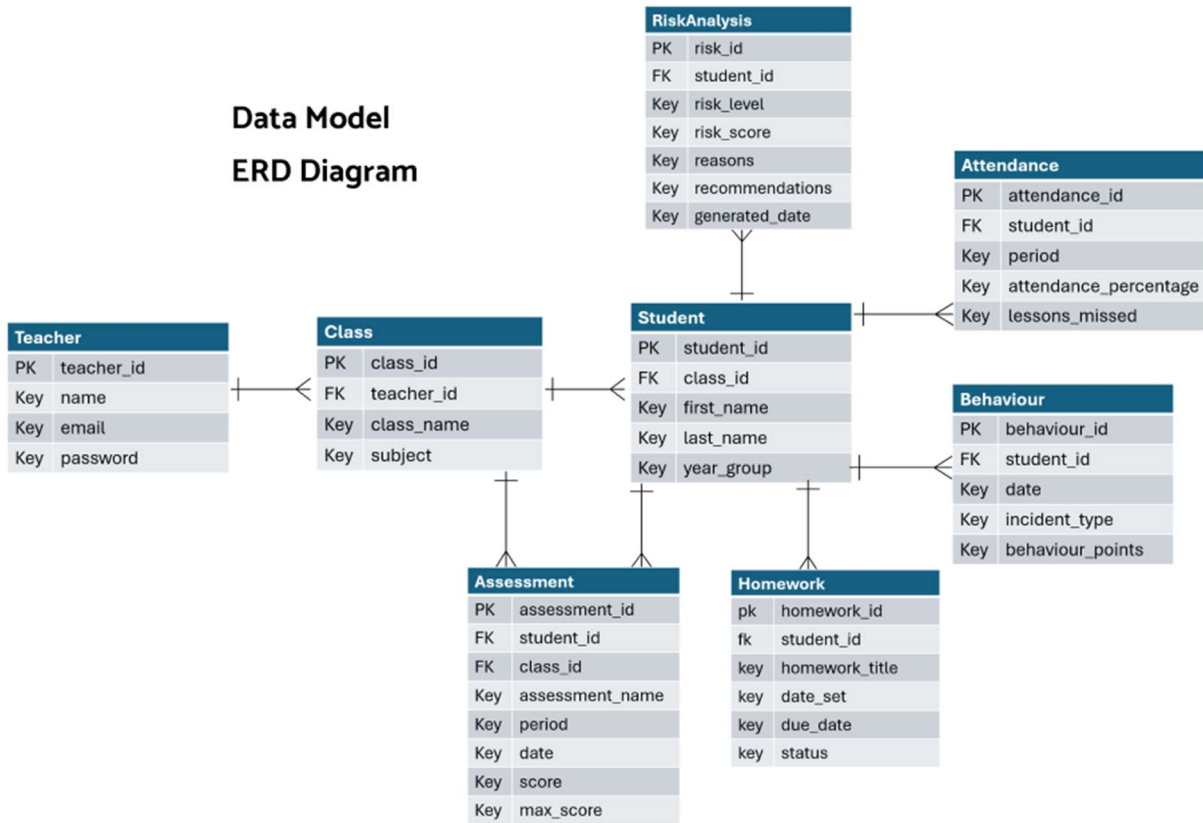


FIGURE 3.2: DATA MODEL ERD DIAGRAM

Figure 3.2 illustrates the relationships between the datasets used within Phronesis. The student entity forms the centre of the data model and is linked to attendance, assessment, behaviour, homework, and risk analysis through the unique student\_id. This allows information from multiple sources to be combined into a single student profile, providing the basis for feature engineering, Random Forest prediction, contextual insights and recommendation generation. Although the prototype uses CSV files rather than a relational database, the ERD represents the intended logical data model for a full implementation of Phronesis. The ERD reflects the original logical design as during implementation, the prediction stage was revised from a stored rule-based risk score to a trained Random Forest model, while behaviour and homework were retained as contextual rather than predictive inputs, this will be shown later on in the project.

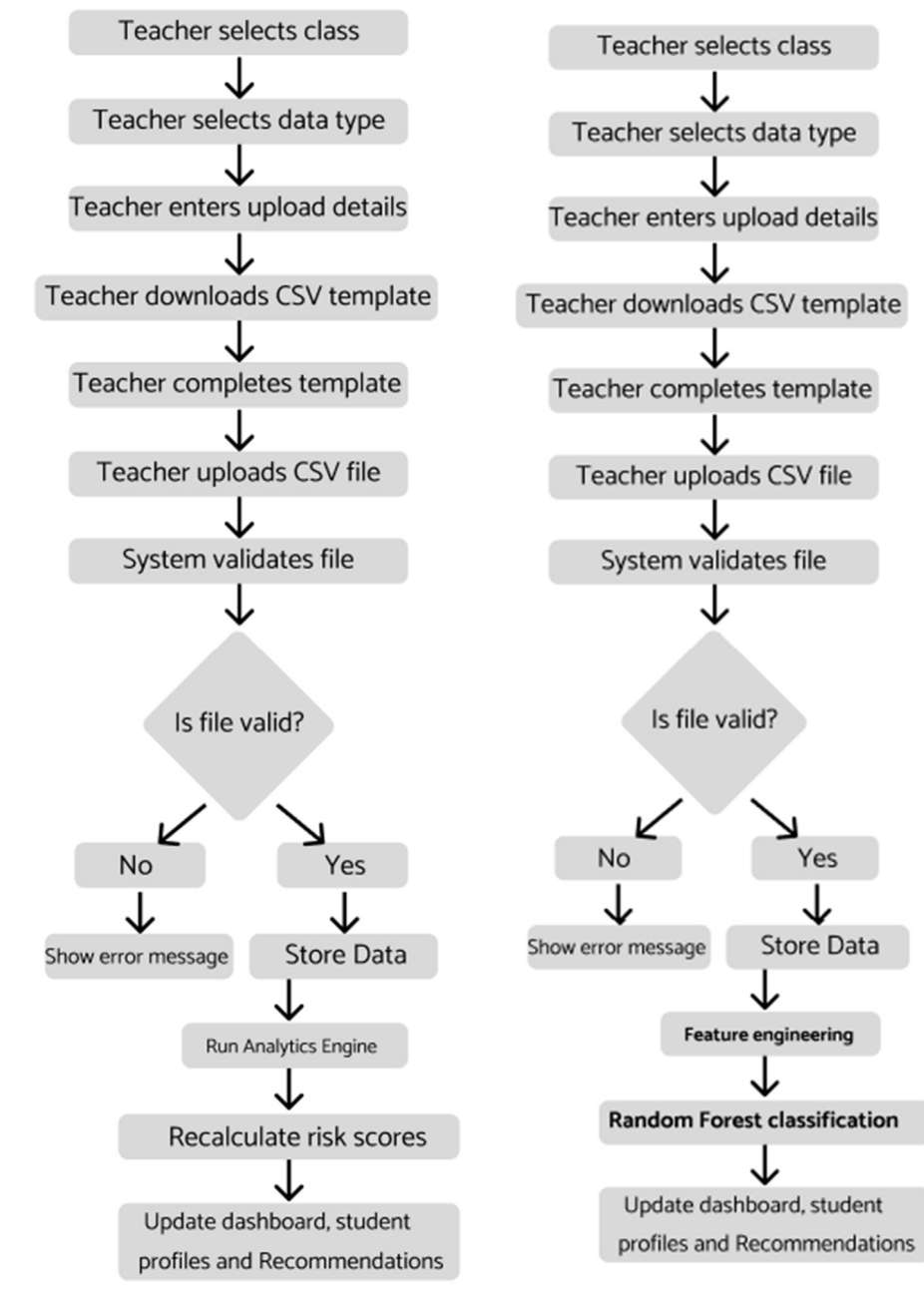


FIGURE 3.3: DATA UPLOAD WORKFLOW

Figure 3.3 shows the original planned upload workflow on the left and on the right the final workflow is provided. Teachers first select a class and data type before uploading CSV files using a standard template. The system validates each file to ensure the required fields are present before processing the data. In the final implementation, the 'Analytics Engine' stage was replaced by feature engineering and Random Forest prediction, after which the dashboard and student views were updated with At Risk, Not At Risk or Insufficient data outputs.

### 3.5 Explainability Engine Design

The original design used a rule-based scoring system to classify students into Low, Medium and High-risk categories as shown in table 3.5.

| Risk Score | Category | Suggested Action                   |
|------------|----------|------------------------------------|
| 0–29       | Low      | Continue monitoring                |
| 30–59      | Medium   | Review student progress            |
| 60–100     | High     | Immediate intervention recommended |

TABLE 3.5: ORIGINAL RISK CATEGORISATION TABLE

During implementation, due to scope, this approach was replaced by a trained Random Forest model following experimentation with several machine-learning classifiers. The final model uses four features: **prev\_failure**, **absences**, **grade\_average** and **grade\_change**. (Further detail is provided in chapter 4: implementation). Where all required features are available, the model classifies a student as either At Risk or Not At Risk. Where required information is missing, the student remains visible within the application but is assigned Insufficient data rather than being forced into a potentially unreliable prediction.

### 3.6 Contextual Insight Design

The Random Forest model generates the academic risk classification, while a separate rule-based layer provides contextual information to support teacher interpretation. This layer examines observable indicators including attendance, assessment average, recent assessment change and previous assessment failure, with behaviour and homework information also displayed where available. These indicators are converted into short, readable statements such as identifying low attendance or declining performance. The contextual insights are designed to help teachers understand the wider situation surrounding a student; however, they are not presented as exact explanations of why the Random Forest produced a particular prediction.

| Risk Factor Identified   | Explanation Generated                                                   |
|--------------------------|-------------------------------------------------------------------------|
| Low attendance           | "Low attendance has contributed to this student's risk level."          |
| Low assessment score     | "Recent assessment performance is below the expected level."            |
| Poor homework completion | "Homework completion has been consistently low."                        |
| Behaviour concerns       | "Repeated behaviour incidents have increased the student's risk score." |

TABLE 3.6: EXAMPLES OF CONTEXTUAL INSIGHTS GENERATED BY PHRONESIS

### 3.7 Recommendation Engine Design

The Recommendation Engine uses the identified risk factors to suggest appropriate evidence-based interventions. Recommendations are generated using predefined rules that map identified risk factors to evidence-based interventions identified in the literature. The recommendations are intended to support, rather than replace, teacher judgement.

| Risk Factor Identified | Recommendation Generated |
|------------------------|--------------------------|
|------------------------|--------------------------|

|                            |                                                            |
|----------------------------|------------------------------------------------------------|
| Low attendance             | Contact parents and monitor attendance.                    |
| Low assessment performance | Provide targeted intervention or one-to-one support.       |
| Poor homework completion   | Review homework expectations and communicate with parents. |
| Behaviour concerns         | Arrange a pastoral meeting and monitor behaviour.          |

**TABLE 3.7: EXAMPLE RECOMMENDATIONS GENERATED BY THE RECOMMENDATION ENGINE**

## 3.8 User Interface Design

### 3.8.1 Initial wireframes

Initial wireframes were produced to explore the overall layout and functionality of the system. The sketches focused on the dashboard, student profile and data upload pages before feedback was gathered from teachers.

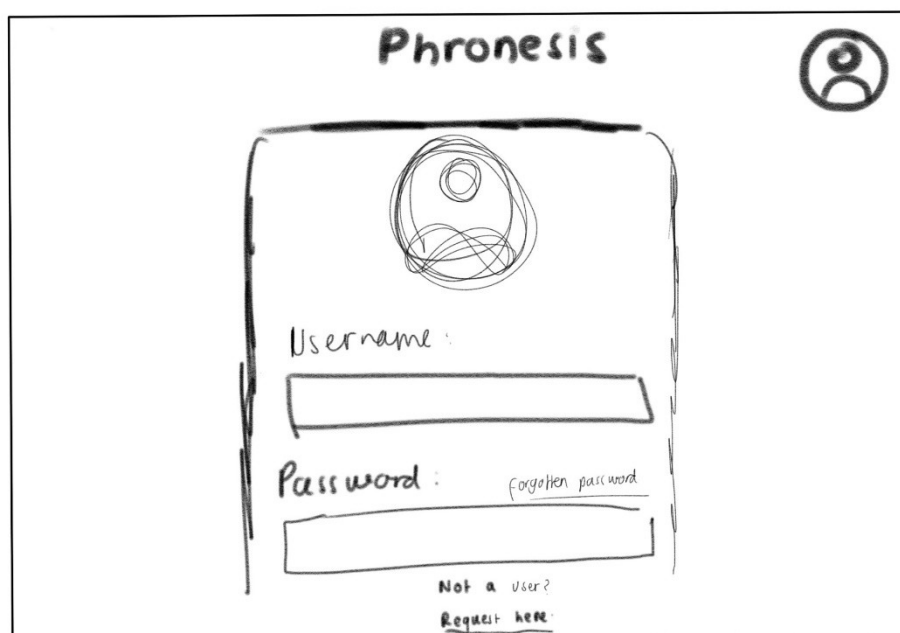


FIGURE 3.4: INITIAL SKETCH OF THE LOGIN PAGE

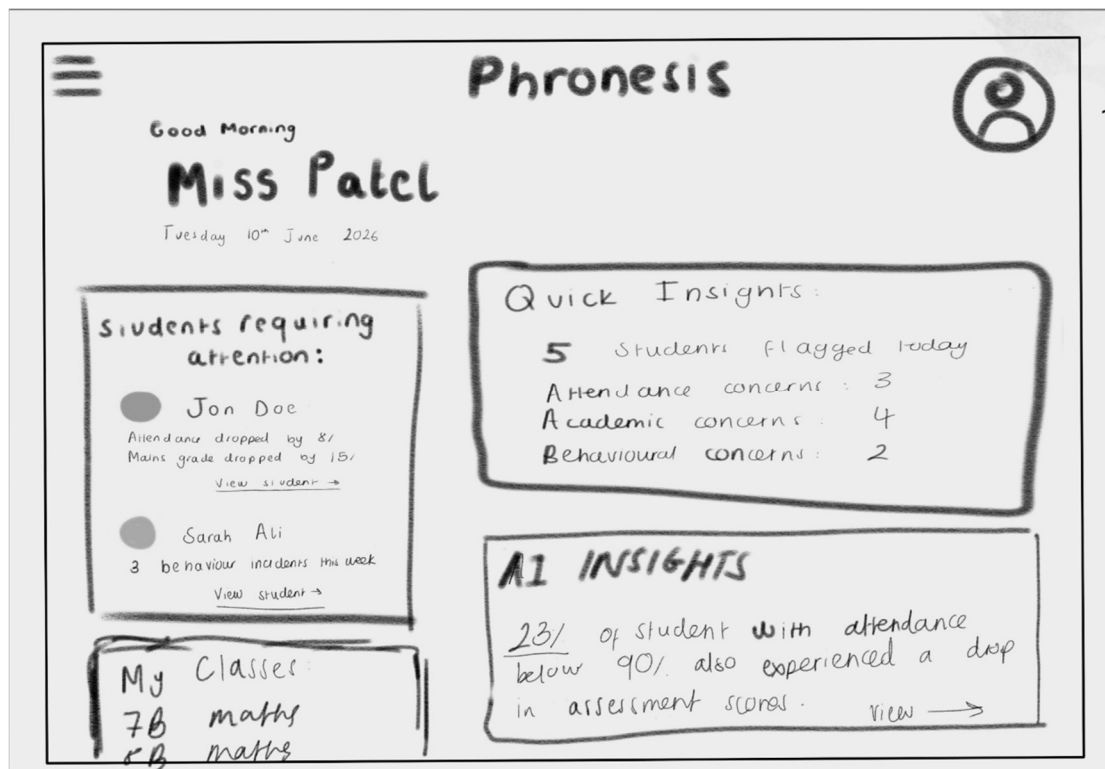


FIGURE 3.5: INITIAL DASHBOARD WIREFRAME SHOWING THE PROPOSED HOMEPAGE LAYOUT.

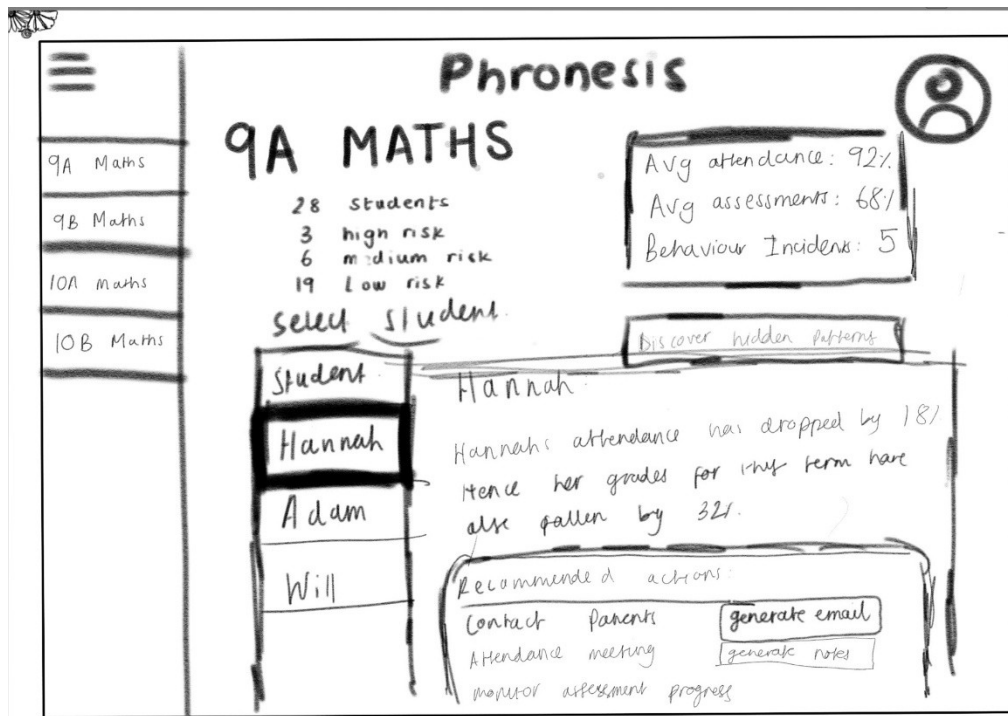


FIGURE 3.6: INITIAL STUDENT PROFILE WIREFRAME ILLUSTRATING THE PLANNED PRESENTATION OF STUDENT INFORMATION.

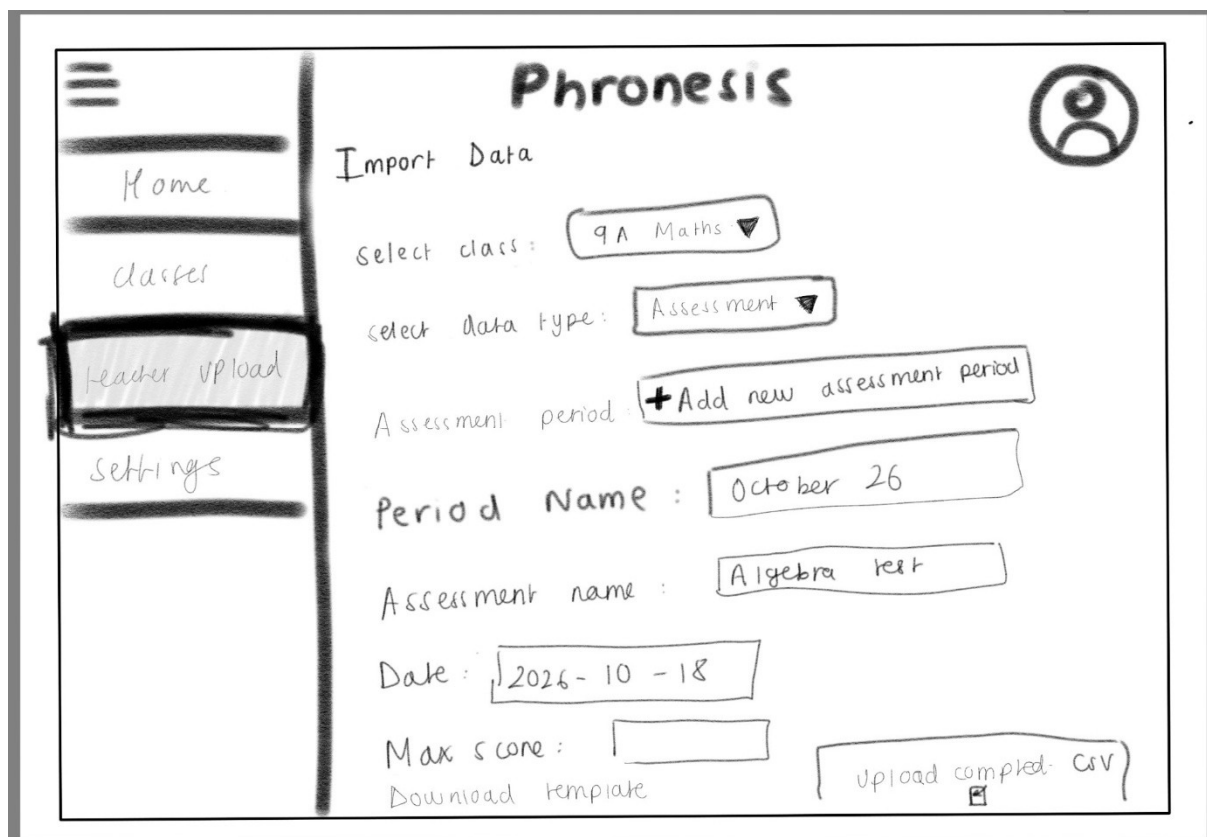
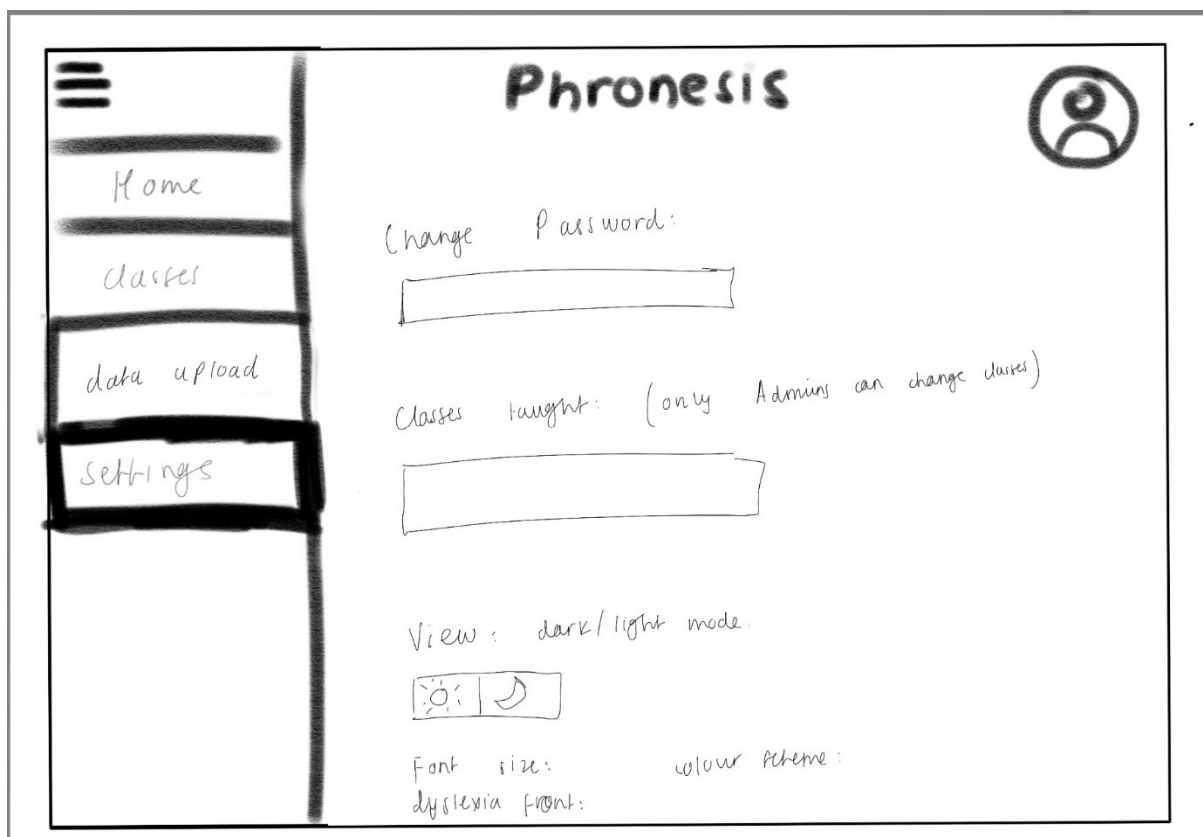


FIGURE 3.7: INITIAL DATA UPLOAD WIREFRAME FOR IMPORTING CSV FILES.



**FIGURE 3.8: INITIAL SETTINGS PAGE**

### 3.8.2 Teacher feedback

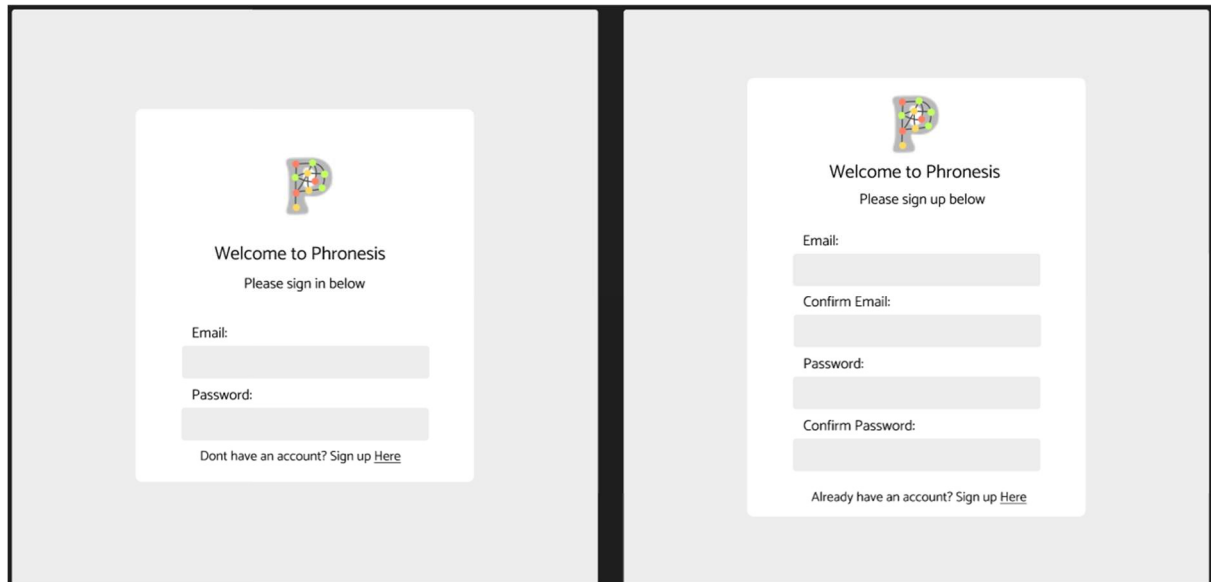
Initial wireframes were presented to teachers from multiple schools to gather feedback before implementation. Their comments informed several design refinements to improve usability and ensure the interface better reflected classroom needs. The main changes made following teacher feedback are summarised in Table 3.8. The original feedback led to colour-coded Low/Medium/High states; during later ML development these were revised to At Risk, Not At Risk and Insufficient data.

| Teacher Feedback                                   | Design Change                                                                          |
|----------------------------------------------------|----------------------------------------------------------------------------------------|
| Logo appeared on every page                        | Moved the logo to the navigation bar to reduce repetition.                             |
| More colour was requested                          | Introduced colour-coded risk indicators for High Risk, Low Risk and insufficient data. |
| Essential information should be easier to identify | Added dashboard summaries and clearer visual hierarchy.                                |

**TABLE 3.8: DESIGN CHANGES MADE FOLLOWING TEACHER FEEDBACK.**

### 3.8.3 Final wireframes

The final wireframes illustrate the completed interface following refinements made throughout the design process. The system includes a dashboard, student profile, data upload pages, and analytical outputs, providing teachers with quick access to key information through a clear and consistent layout. The final design forms the basis of the prototype presented in the following chapter.



**FIGURE 3.9: FINAL LOGIN PAGE WIREFRAME INCORPORATING TEACHER FEEDBACK AND IMPROVED NAVIGATION**

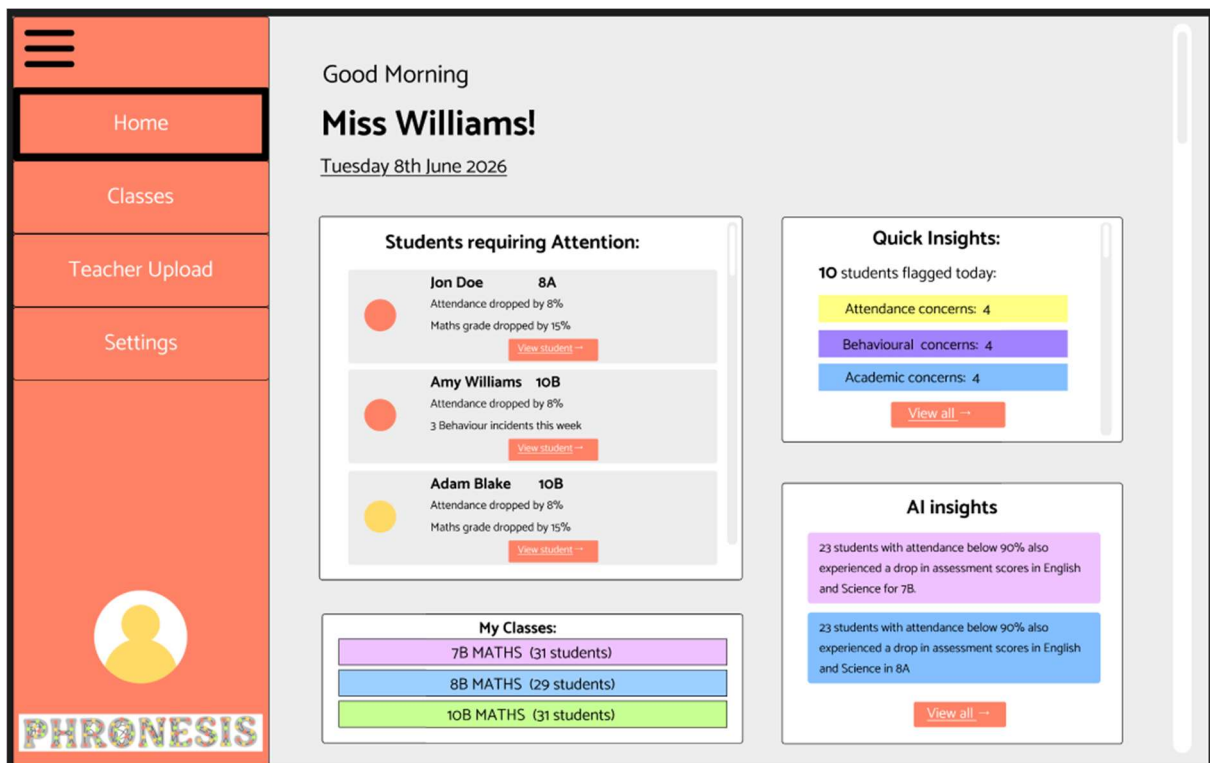


FIGURE 3.10: FINAL DASHBOARD WIREFRAME DISPLAYING STUDENT SUMMARIES, ALERTS AND COLOUR-CODED RISK INDICATORS.

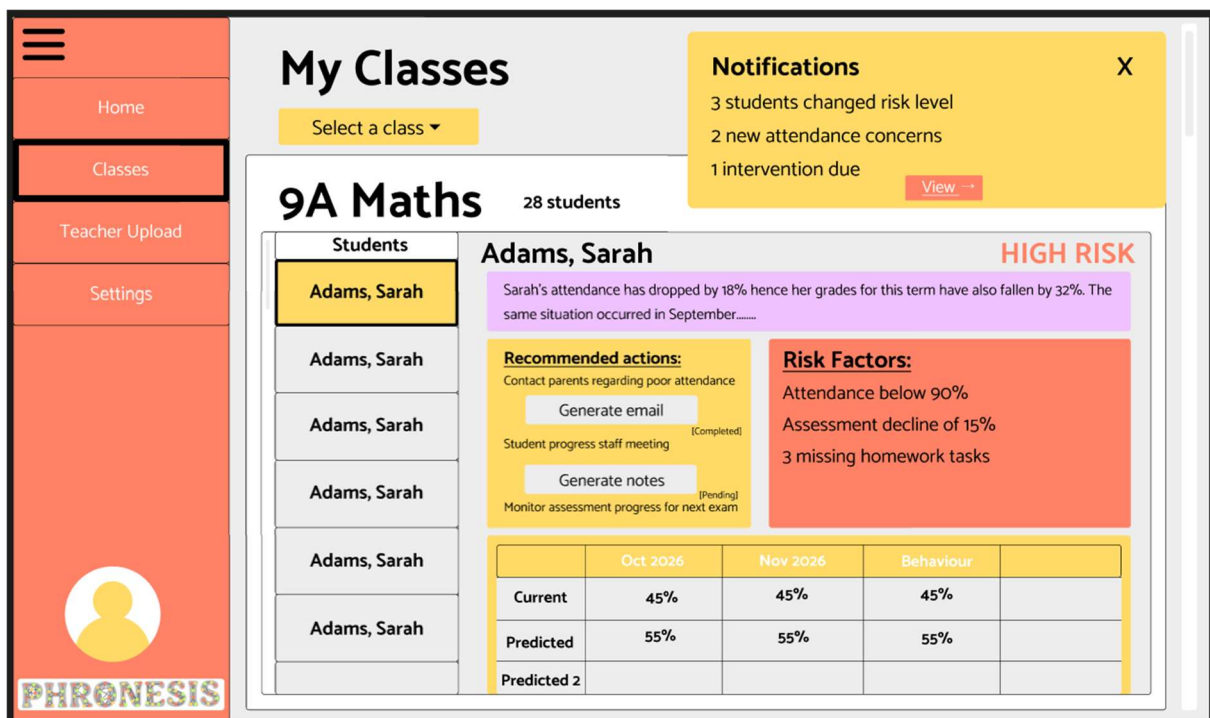


FIGURE 3.11: FINAL STUDENT PROFILE WIREFRAME SHOWING INTEGRATED ATTENDANCE, ASSESSMENT, BEHAVIOUR AND HOMEWORK INFORMATION.

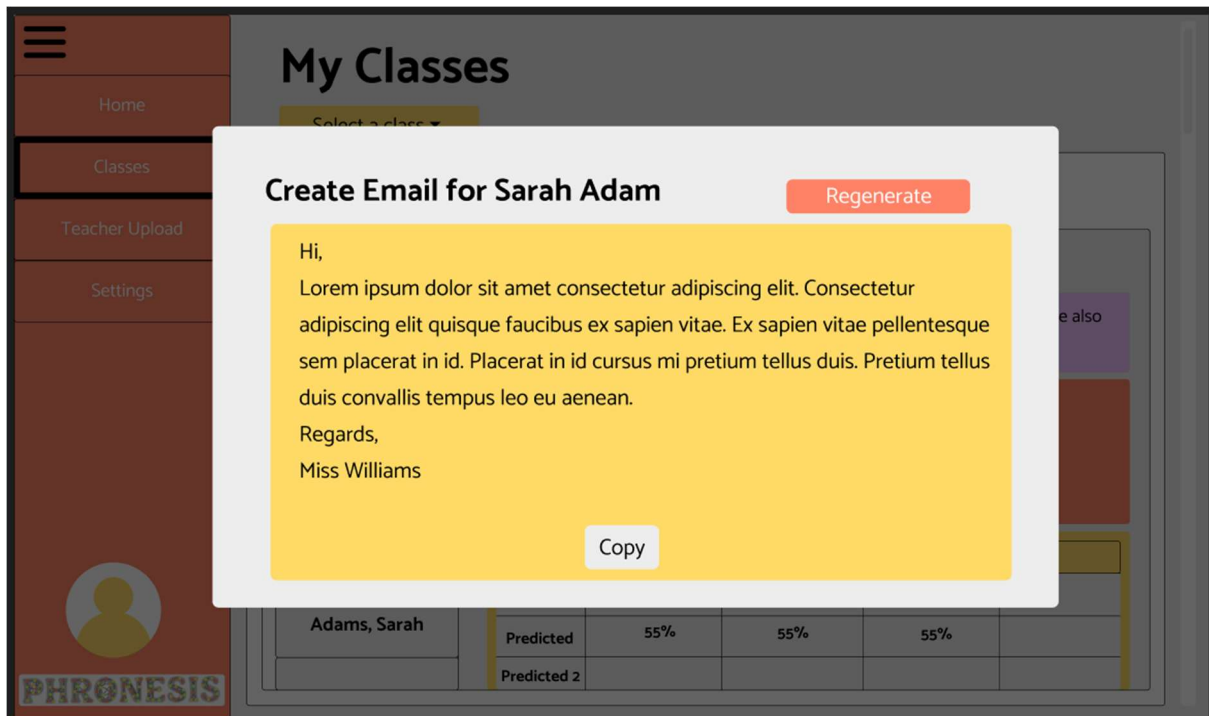


FIGURE 3.12: EMAIL GENERATOR SCREEN

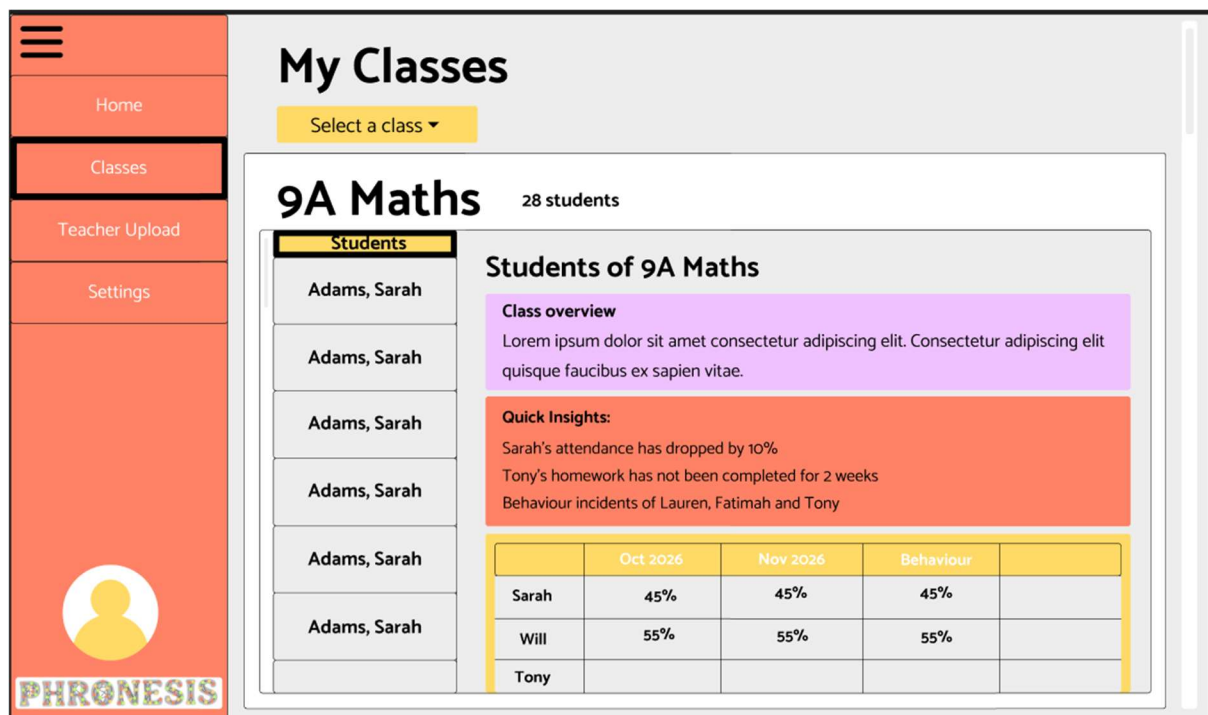


FIGURE 3.13: FINAL CLASS PROFILE WIREFRAME SHOWING GENERAL INTEGRATED ATTENDANCE, ASSESSMENT, BEHAVIOUR AND HOMEWORK INFORMATION.

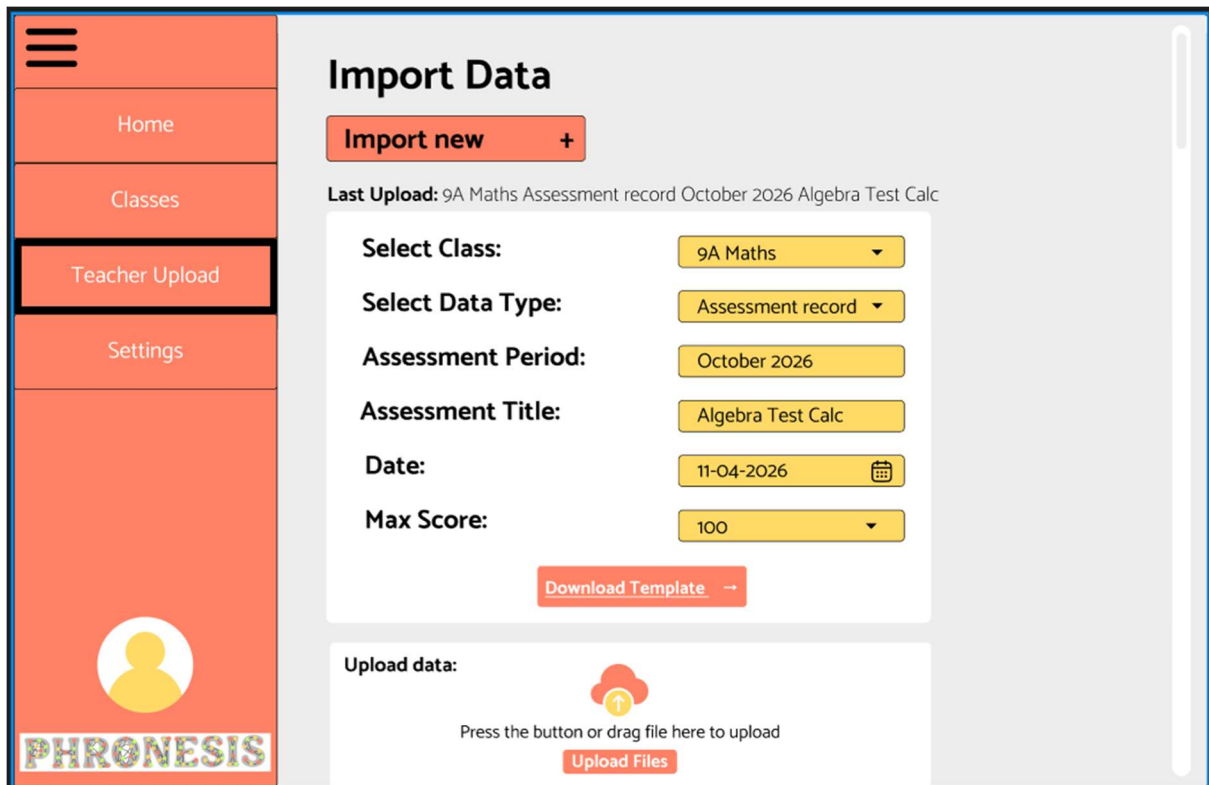


FIGURE 3.14: FINAL DATA UPLOAD WIREFRAME ILLUSTRATING THE CSV IMPORT PROCESS.

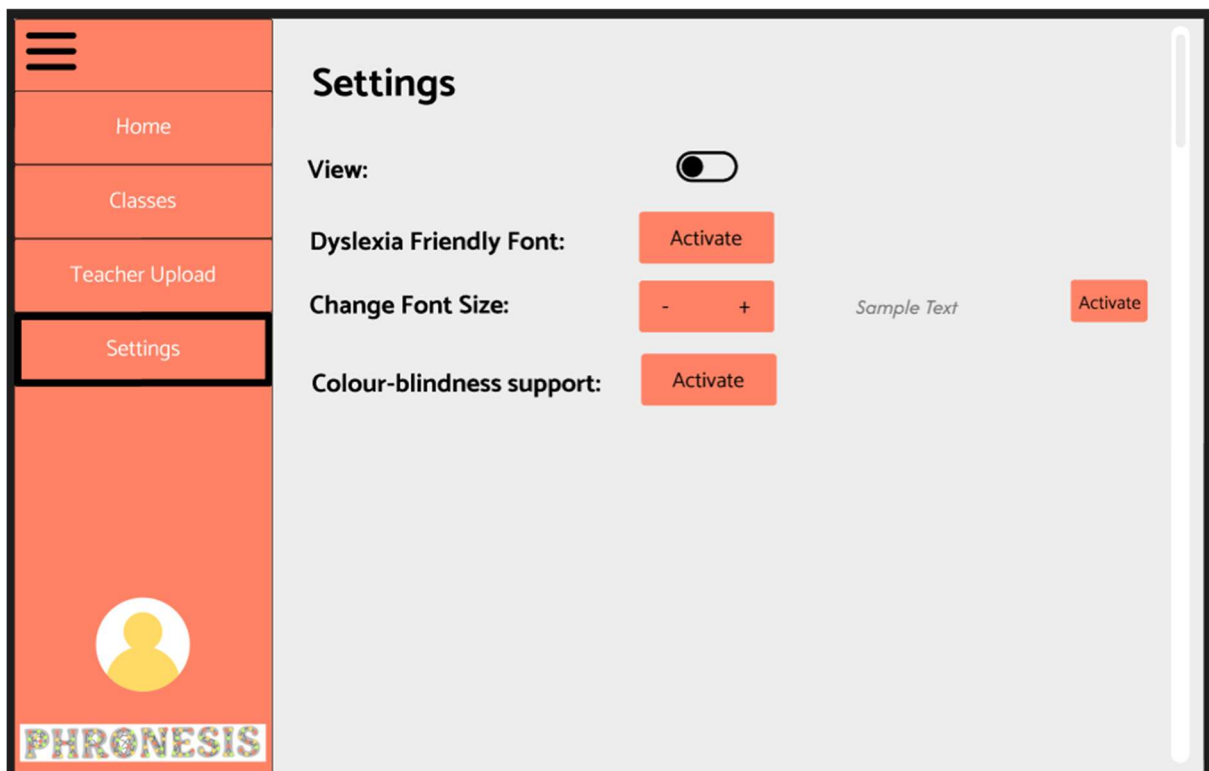
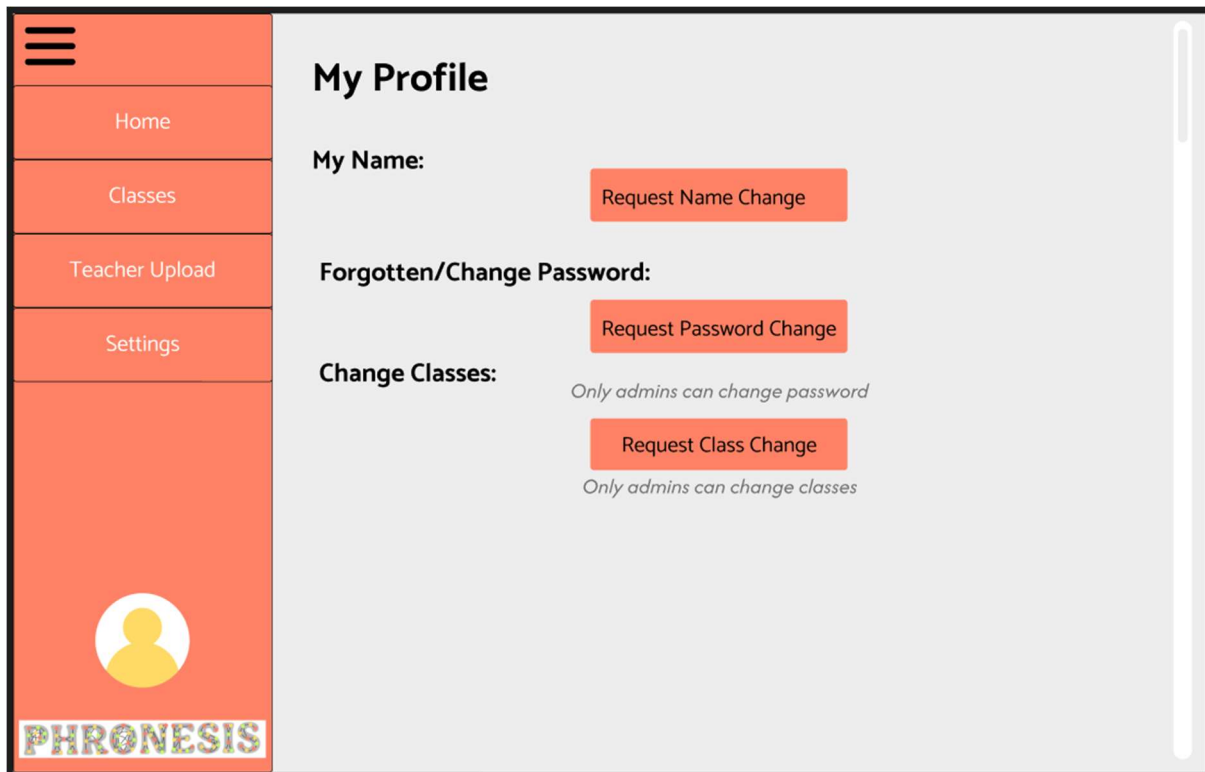


FIGURE 3.15: FINAL SETTINGS PAGE WIREFRAME.



**FIGURE 3.14: FINAL PROFILE PAGE WIREFRAME.**

#### *3.8.4 Design Justification*

The interface design was informed by findings from the literature review, teacher interviews, and project survey. Feedback from teachers influenced several design decisions, including the use of colour-coded risk indicators, simplified navigation and reduced visual repetition. These changes aimed to improve usability while supporting faster interpretation of student data and more informed decision making.

### **3.9 Work Plan**

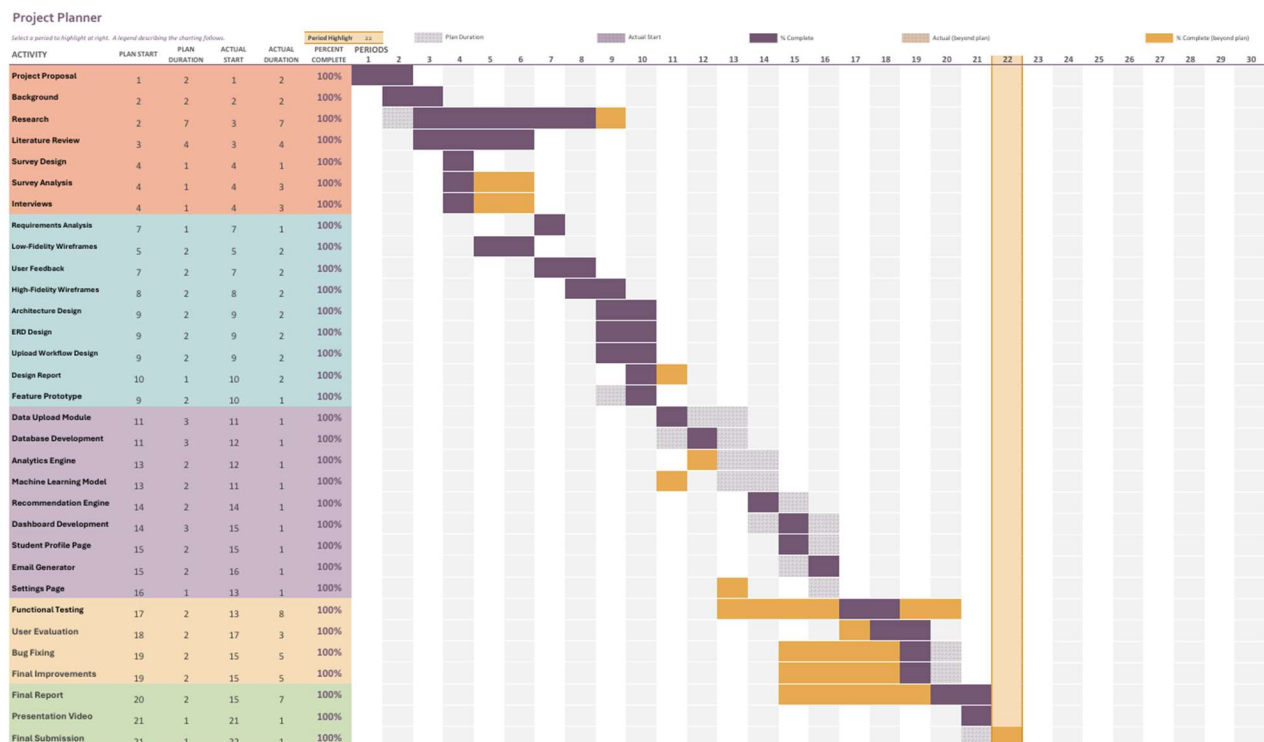
#### *3.9.1 Development Methodology*

An iterative development approach was adopted, allowing features to be implemented, evaluated and refined throughout the project in response to teacher feedback.

#### *3.9.2 Gantt Chart*

The overall project schedule, including research, design, implementation, testing, and report writing, is shown in Figure 3.15.

**FIGURE 3.15: GANTT CHART SHOWING THE PLANNED DEVELOPMENT SCHEDULE FOR PHRONESIS.**



### 3.9.3 Milestones

Table 3.9 below presents key milestones for this project.

| Milestone              | Outcome                                                                      |
|------------------------|------------------------------------------------------------------------------|
| Literature Review      | Completed critical review and identified the research gap.                   |
| System Design          | Produced the architecture, data model and interface design.                  |
| Prototype Development  | Implemented the core risk analysis, explainability ,recommendation features. |
| Testing and Evaluation | Validated the prototype using sample datasets and teacher feedback.          |
| Final Report           | Completed documentation and project submission.                              |

**TABLE 3.9: PROJECT MILESTONES.**

## 3.10 Evaluation Strategy

Phronesis was evaluated using a combination of technical and user-based methods. The machine-learning model was assessed using accuracy, precision, recall, F1, ROC-AUC, Average Precision and final holdout testing. Functional and integration testing was used to check the complete application workflow, alongside incomplete-data and edge-case testing. User evaluation involved a task-based questionnaire, with 13 consenting responses included in the final analysis, followed by three interviews to gather more detailed feedback. Accessibility and usability feedback was also considered. The full evaluation process and results are presented in Chapter 5.

## Chapter 4: Implementation (1915 words)

### 4.1 Implementation Overview

The preliminary report demonstrated the basic workflow of Phronesis, whereby a teacher will upload a class, student, subject, assessment and attendance data. This was then combined to generate risk predictions for students, explanations and tailored recommendations for next steps. The initial prototype contained rule-based risk predictions that were manually defined, rather than using a trained ML model.

In the final development, the prototype was expanded significantly. The rule-based prototype was replaced with a trained Random Forest model. The app also was adapted into a complete Streamlit application with full user workflow.

Teacher data is uploaded, stored and processed. Key variables are converted into the model features where model interprets them to generate risk predictions. Final outputs were revised from High, Medium and Low risk to At Risk, Not At Risk and Insufficient data.

### 4.2 Development Environment and Final System Architecture

The app was developed in Python using scikit-learn for the ML and Pandas to process the data. The model was loaded and saved through joblib and Streamlit allowed the implementation of the web application. Git/GitHub provided version control.

The final system was separated into modular components: The user interface, data-management modules, data-processing modules, machine learning model and styling. Each module handled one or a few functions which made it easier to manage independently and maintain. Figure 4.1 indicates different files and folders needed for the projects to allow for this separation of concerns.

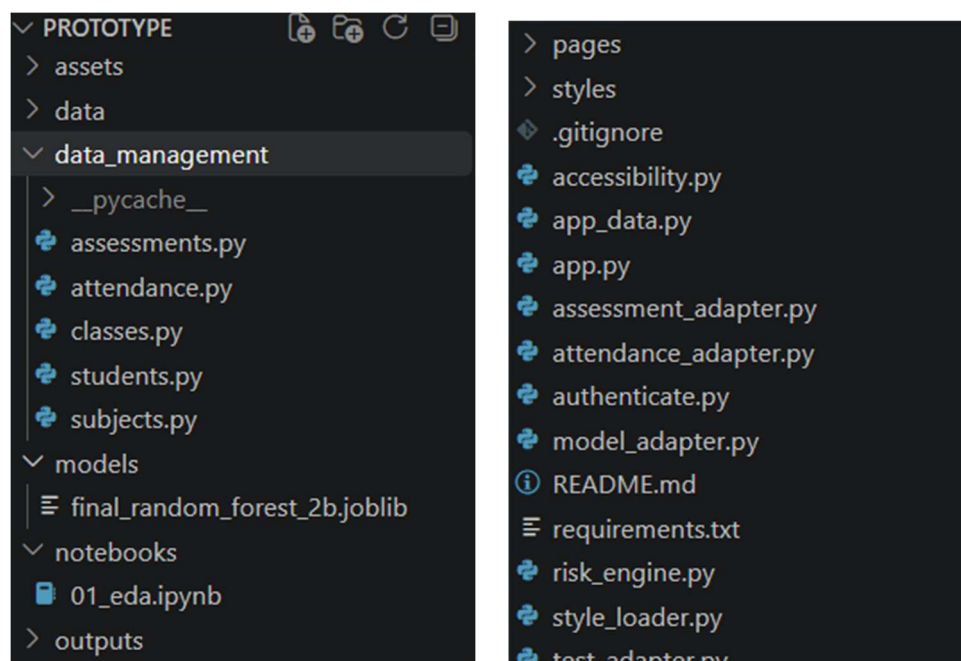


FIGURE 4.1- FILES SHOWING THE FINAL MODULAR PROJECT STRUCTURE

load\_app\_data() is the central pipeline that loaded the current teacher's data, processed assessments and attendance information, merges the datasets, derives the model feature information, applies predictions and finally returns one combined dataset that the interface can use. This idea was introduced during app development so that different Streamlit pages did not independently repeat the same backend operations.

The general workflow is denoted by the diagram below:



FIGURE 4.2- HIGH LEVEL ARCHITECTURE FLOW DIAGRAM

## 4.3 Machine Learning Implementation

### 4.3.1 Model Development and Selection (250)

Several models were tested before coming to the final Random Forest conclusion: Majority-class baseline, Logistic Regression, Decision Tree and Random Forest models. First, the risk classification was finalised based on the final grade "G3". Students with a G3 < 10 were labelled as At Risk, and students with a G3 >= 10 were Not At Risk (as shown in figure 4.3). Multiple feature experiments were also completed and classifiers were compared.

```
#remove column first if created already
if "at_risk" in dfCsv.columns:
    dfCsv.drop(columns=["at_risk"], inplace=True)

#insert() to add a column at pos 33 where value is 1 if g3<10, else 0
dfCsv.insert(33, "at_risk", (dfCsv["G3"] < 10).astype(int))

#count how many students at risk/not at risk
dfCsv["at_risk"].value_counts()
#dfCsv.head()
```

```
at_risk
0      549
1      100
Name: count, dtype: int64
```

FIGURE 4.1 CODE SNIPPET OF THE CREATION OF THE BINARY TARGET VARIABLE USED DURING MODEL TRAINING. STUDENTS WITH G3 BELOW 10 WERE LABELLED AS AT RISK

After initial tests, it was found that the dataset was imbalanced (Shown in figure 4.3.2), whereby the majority baseline achieved an 84.62% accuracy but 0% precision, recall and F1 for At Risk classification. This was strong evidence that accuracy alone would be a misleading metric to measure the performance of the prediction models.

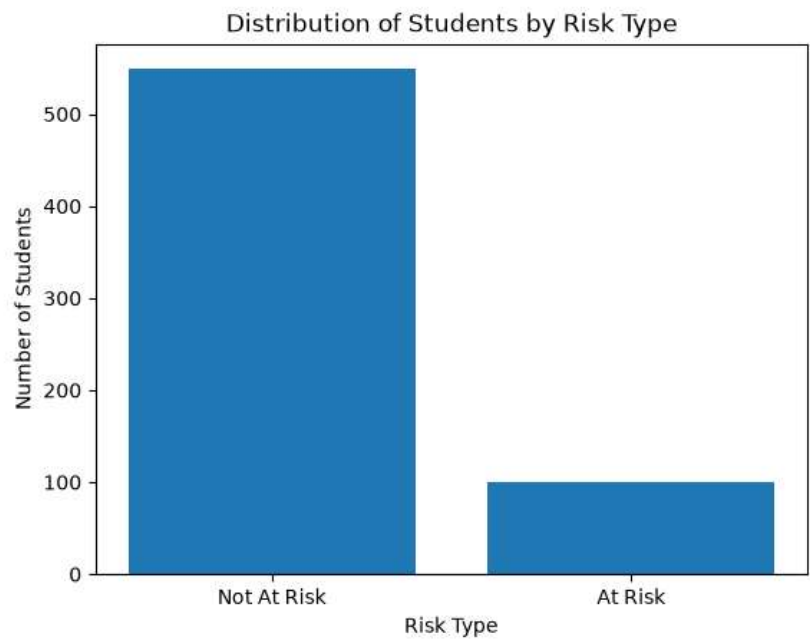


FIGURE 4.3.2- DATASET DISTRIBUTION BY RISK TYPE

The final model candidates, Logistic 2B @ 0.40, Depth-2 Tree 2B and Tuned Random Forest 2B, were compared with results shown in Figure 4 and 5. The tuned Random Forest achieved the highest F1 score at 82.35%, narrowly exceeding the Depth-2 Decision Tree at 82.08%. It also maintained high recall and strong ROC-AUC performance, providing the strongest overall balance across the main evaluation metrics.

| Model                         | Accuracy           | Precision          | Recall | F1                 | ROC-AUC            | Average Precision  | Predicted At Risk |
|-------------------------------|--------------------|--------------------|--------|--------------------|--------------------|--------------------|-------------------|
| Logistic Regression 2B @ 0.40 | 0.9364161849710982 | 0.8051948051948052 | 0.775  | 0.7898089171974523 | 0.9729498861047836 | 0.8694382490204141 | 77                |
| Decision Tree 2B depth 2      | 0.9402697495183044 | 0.7634408602150538 | 0.8875 | 0.8208092485549133 | 0.9615888382687927 | 0.8138715328107275 | 93                |
| Tuned Random Forest 2B        | 0.9421965317919075 | 0.7777777777777778 | 0.875  | 0.8235294117647058 | 0.9665148063781321 | 0.8584478519181583 | 90                |

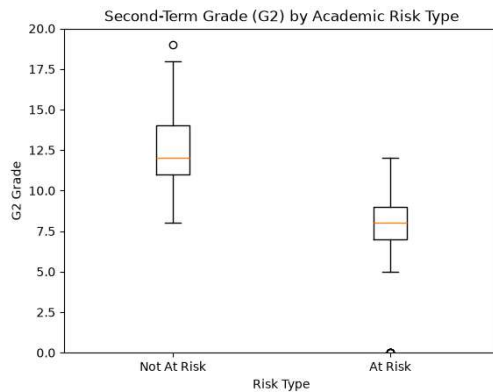
FIGURE 4.3.3 - FINAL MODEL COMPARISON CSV

| Model | Accuracy                      | Precision          | Recall             | F1     | ROC-AUC            | Average Precision  | Predicted At Risk |
|-------|-------------------------------|--------------------|--------------------|--------|--------------------|--------------------|-------------------|
| 1     | Logistic Regression 2B @ 0.40 | 0.9364161849710982 | 0.8051948051948052 | 0.775  | 0.7898089171974523 | 0.9729498861047836 | 77                |
| 2     | Decision Tree 2B depth 2      | 0.9402697495183044 | 0.7634408602150538 | 0.8875 | 0.8208092485549133 | 0.9615888382687927 | 93                |
| 3     | Tuned Random Forest 2B        | 0.9421965317919075 | 0.7777777777777778 | 0.875  | 0.8235294117647058 | 0.9665148063781321 | 90                |

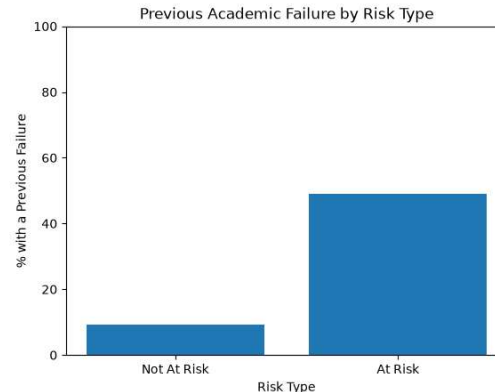
FIGURE 4.3.4 FINAL MODEL COMPARISON TABLE

#### 4.3.2 Final Model Features (150)

The initial dataset contained many variables. And so, experimentations were carried out to identify which features were most useful to predict academic risk. As the dataset already contained the final academic outcome (G3- final grade) the relationships between different variables and student risk were investigated. Visual analysis, including box plots a strong relationship between previous assessment performance, previous failures and final academic risk (as shown in Figure 4.3.5 and 4.3.6).



**4.3.5 SECOND TERM GRADES G2 IN BOTH RISK CATEGORIES**



**4.3.6 PREVIOUS ACADEMIC FAILURE BY RISK TYPE**

The final model used four main features (also shown in code snippet below):

- prev\_failure: Whether the student had previously failed an assessment in that subject.
- absences: The total sessions minus sessions attended.
- grade\_average: Recent assessment performance converted to the models internal 0-20 scale.
- grade\_change: The direction of recent assessment performance.

```
#get latest two assessments
latest_two = valid_assessments.tail(2)
previous_grade = float(latest_two["model_grade"].iloc[0])
latest_grade = float(latest_two["model_grade"].iloc[1])

#average grade
grade_average = (previous_grade + latest_grade)/2

#grade change
grade_change = latest_grade - previous_grade

#get all assessments before latest one
previous_assessments = valid_assessments.iloc[:-1]
```

**CODE SNIPPET 1: CALCULATION OF GRADE\_AVERAGE AND GRADE\_CHANGE FROM THE TWO**

Homework and behaviour are not model inputs; however, they are shown separately to provide contextual indicators for teacher interpretation.

### 4.3.3 Feature Engineering and Subject-Specific Processing (200)

The raw school data inputted by users contains values such as score, max\_score, pass\_mark, date, student and class. Attendance data contains total\_sessions and sessions\_attended. This data required transforming before being inputted into the prediction model.

Assessment records were validated, with scores being converted into a consistent scale and records being sorted by date. The most recent assessments are used to calculate new variables and are treated as the equivalent recent-grade inputs when calculating grade\_average and grade\_change.

grade\_average:  $(G1+G2)/2$  (G2=Most recent assessment results, G1= Second most recent results)

grade\_change:  $G2-G1$

prev\_failure: Whether a previous assessment was failed relative to its pass\_mark

absences: total\_sessions – sessions\_attended

Assessment history is grouped by both student\_id and subject. For example, this prevents a Maths result being treated as part of a Spanish grade trend. This is important because students can perform differently across subjects and so predictions are kept as subject-specific rather than overall. This transformation therefore converts teacher-entered data into the exact feature format required by the Random Forest model.

## 4.4 Handling Incomplete and Missing Data

It is common within a school environment for data to remain incomplete, students may have missing attendances or assessment data. New students or absent students may have missing attendance or little assessment history, forcing a prediction from incomplete inputs to be misleading.

Phronesis' system checks whether all the required model features are available, with only complete student rows being passed into the Random Forest model. The incomplete students remain visible in the application, but they receive the category title of "Insufficient data" (as shown in figure 4.4.1 and code shown in figure 4.4.2). This avoids any missing values being replaced with arbitrary zeros, so no fabricated risk results are given.

| Students                      |               |              |                      |                     |                   |  |
|-------------------------------|---------------|--------------|----------------------|---------------------|-------------------|--|
| Student ID                    | Student       | Attendance % | Assessment Average % | Assessment Change % | Risk              |  |
| <input type="checkbox"/> S015 | Aisha Hussain | 82.57429     | 41.500000            | -5.000000           | At Risk           |  |
| <input type="checkbox"/> S006 | Oliver Green  | 90.85743     | 39.000000            | -8.000000           | At Risk           |  |
| <input type="checkbox"/> S002 | Yusuf Khan    | 93.43871     | 42.000000            | -8.000000           | At Risk           |  |
| <input type="checkbox"/> S012 | Adam Rahman   | None         | 44.500000            | -7.000000           | Insufficient data |  |
| <input type="checkbox"/> S003 | Chloe Bennett | None         | 72.000000            | 4.000000            | Insufficient data |  |
| <input type="checkbox"/> S008 | Ethan Clarke  | None         | 68.500000            | 3.000000            | Insufficient data |  |

**FIGURE 4.4.1 DEMONSTRATES INSUFFICIENT DATA RISK CATEGORY BASED ON LACK OF ATTENDANCE DATA**

```

#assessment and attendance data are valid?
valid_status =(
    (merged_features["assessment_status"] == "Valid")
    &
    (merged_features["attendance_status"] == "Valid")
)

#start with a default-insufficient data
merged_features["prediction_status"] = "Insufficient data"

#valid and complete- are ready for prediction
ready_rows = all_features_present & valid_status

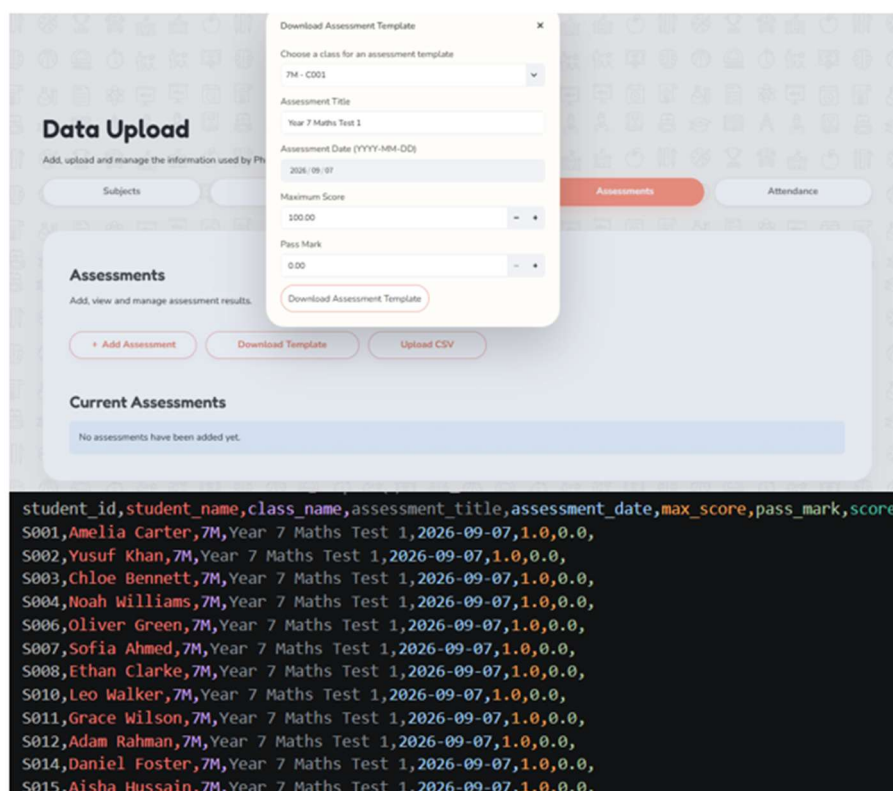
```

**FIGURE 4.4.2 CODE WHICH SETS ALL DATA TO INSUFFICIENT UNLESS FILLED OTHERWISE**

The demonstration dataset intentionally included students with missing attendance or insufficient assessment history so that this incomplete-data pathway could be tested. This provides a safer and more transparent way of warning teachers that additional information is required before a reliable prediction can be generated.

## 4.5 Data Management and Validation

As a teacher, through the application, it is possible to manage subjects they teach, classes, students, assessment and attendance records. Teachers have the option to manually input the data for a single student on the application (as shown in Figure 4.5.1 and 4.5.2) or download a pre-filled CSV template for bulk data input (As shown in figure 4.5.3). Existing records can also be edited and deleted (as shown in figure 4.5.4).



**FIGURE 4.5.1 DEMONSTRATES HOW USER CAN DOWNLOAD AN ASSESSMENT TEMPLATE PREFILLED WITH ASSESSMENT DATA. RESULTING IN THE TEACHER**

```

####MAKE TEMPLATE WITH STUDENT ID,NAME###
assessment_template =templ_students[["student_id","name"]].copy()
assessment_template =assessment_template.rename(columns={"name":"student_name"})

#CLASS NAME PREFILLED
assessment_template["class_name"] = templ_selected_class_name
assessment_template["assessment_title"] = assessment_title
assessment_template["assessment_date"] = assessment_date
assessment_template["max_score"] = max_score
assessment_template["pass_mark"] = pass_mark

#teacher only needs to fill this
assessment_template["score"] = ""

#template into csv
assessment_template_csv =assessment_template.to_csv(index=False)

```

FIGURE 4.5.2 CODE HANDLES PREFILLED CSV 1

**Add Assessment**

Enter the assessment result below.

Class: 7M - C001

Subject: Maths

Student: Amelia Carter - S001

Assessment Title:

Assessment Date: 2026/09/27

Maximum Score: 1.00

Student Score: 0.00

Pass Mark: 0.00

**Add Assessment**

FIGURE 4.5.3 ADD ONE STUDENTS ASSESSMENT RESULT

**Edit Assessment**

Update Year 7 Maths Baseline Test.

Assessment Title: Year 7 Maths Baseline Test

Assessment Date: 2026/09/07

Maximum Score: 1.00

Student Score: 0.83

Pass Mark: 0.50

**Cancel** **Save Changes**

| Student ID | Student Name  | Subject | Assessment Title           | Assessment Date | Score | Max Score | Pass M |
|------------|---------------|---------|----------------------------|-----------------|-------|-----------|--------|
| S001       | Amelia Carter | Maths   | Year 7 Maths Baseline Test | 2026-09-07      | 0.83  | 1         |        |
| S002       | Oliver Green  | Maths   | Year 7 Maths Number Test   | 2026-09-14      | 0.46  | 1         |        |
| S003       | Sofia Ahmed   | Maths   | Year 7 Maths Number Test   | 2026-09-14      | 0.78  | 1         |        |
| S004       | Ethan Clarke  | Maths   | Year 7 Maths Number Test   | 2026-09-14      | 0.46  | 1         |        |
| S005       | Leo Walker    | Maths   | Year 7 Maths Number Test   | 2026-09-14      | 0.7   | 1         |        |
| S006       | Noah Williams | Maths   | Year 7 Maths Number Test   | 2026-09-14      | 0.79  | 1         |        |
| S007       | Oliver Green  | Maths   | Year 7 Maths Number Test   | 2026-09-14      | 0.43  | 1         |        |
| S008       | Sofia Ahmed   | Maths   | Year 7 Maths Number Test   | 2026-09-14      | 0.68  | 1         |        |
| S009       | Ethan Clarke  | Maths   | Year 7 Maths Number Test   | 2026-09-14      | 0.67  | 1         |        |
| S010       | Leo Walker    | Maths   | Year 7 Maths Number Test   | 2026-09-14      | 0.74  | 1         |        |

Selected: Year 7 Maths Baseline Test - Daniel Foster **Edit** **Delete**

FIGURE 4.5.4 EDIT/DELETE STUDENT ASSESSMENT 1

Before uploaded data was added to the application data, it was necessary for the data to be validated as bad input could create invalid model features or misleading predictions. Some examples of validation used in Phronesis include required columns, missing values, valid student\_ids, whole-number attendance sessions, sessions\_attended <= total\_sessions, score/pass\_mark consistency. If entered incorrectly then specific error

messages were shown (as shown in figure 4.5.5). Each teacher account uses separate student data; this ensures accounts do not mix with one another using the `get_user_file()` (as implemented in figure 4.5.6)

```
#invalid date?-flag error
invalid_dates =uploaded_assessments["assessment_date"].isna().any()

if invalid_dates:
    st.error("One or more assessment dates are invalid.")
    return

#empty values?-flag error
if empty_values.any():
    st.error("Student ID, class name and assessment title cannot be empty.")
    return

#missing columns list- if any needed columns missing-flag error
missing_columns =[]
for column in required_columns:
    if column not in uploaded_assessments.columns:
        missing_columns.append(column)

#missing columns?-flag error
if missing_columns:
    st.error(f"Missing required columns: {missing_columns}")
    return

if invalid_numbers:
    st.error("Score, maximum score and pass mark must contain numbers.")
    return

#score cant be negative
negative_score =(uploaded_assessments["score"] <0).any()

if negative_score:
    st.error("Student score cannot be negative.")
    return

# FIX! score cant be higher than max score
score_above_maximum =(uploaded_assessments["score"] >
                        uploaded_assessments["max_score"]).any()

if score_above_maximum:
    st.error("Student score cannot be greater than maximum score.")
    return
```

FIGURE 4.5.5 SNIPPET OF ERROR HANDLING CODE IN ASSESSMENTS.PY

```
def create_user_files():
    folder =get_user_folder()

    #csv files and their columns
    files ={
        "students.csv":["student_id","name","year_group"],
        "subjects.csv":["subject_id","subject_name"],
        "classes.csv":["class_id","class_name","subject_id","year_group"],
        "class_students.csv":["class_id","student_id"],
        "assessment.csv":["student_id","subject","assessment_title","assessment_date","score","max_score","pass_mark"],
        "attendance.csv":["student_id","total_sessions","sessions_attended"],
        "homework_completion.csv":["student_id","homework_completion"],
        "behaviour.csv":["student_id","behaviour_incidents"]
    }
```

FIGURE 4.5.6 GET USER FILES FUNCTION DESIGNED TO RECEIVE THE LOGGED IN USERS FILES ONLY

## 4.6 Final Application Interface and Decision-Support Features

### 4.6.1 Dashboard (130)

The dashboard provides the first overview screen once logged in. Its aim is to provide a quick summary of their classes and an overview on student risk information. This is carried out by combining the processed school data and model outputs from the central `load_app_data()` pipeline rather than calculating risk separately on the page (as shown in figure 4.6).

```
def load_app_data():
    #####READ DATA#####
    students =pd.read_csv(get_user_file("students.csv"))
    assessment =pd.read_csv(get_user_file("assessment.csv"))
    attendance =pd.read_csv(get_user_file("attendance.csv"))
    classes =pd.read_csv(get_user_file("classes.csv"))
    subjects =pd.read_csv(get_user_file("subjects.csv"))
    class_students =pd.read_csv(get_user_file("class_students.csv"))

    #homework data might not exist yet
    homework_file =get_user_file("homework_completion.csv")
    if os.path.exists(homework_file):
        homework =pd.read_csv(homework_file)
    else:
        homework =pd.DataFrame(columns=["student_id","homework_completion"])

    #same for behaviour data
    behaviour_file =get_user_file("behaviour.csv")
    if os.path.exists(behaviour_file):
        behaviour =pd.read_csv(behaviour_file)
    else:
        behaviour =pd.DataFrame(columns=["student_id","behaviour_incidents"])

    #####MODEL DATA#####
    assessment_features =process_assessments(assessment)
    attendance_features =process_attendances(attendance)
    merged_features =merge_model_features(students,assessment_features,attendance_features)
    prediction_results =risk_predict(merged_features)

    #####CLASS DATA#####
    #merge class_students with classes, subjects, and students to get complete class data
    class_data =class_students.merge(classes,on="class_id",how="left")
    class_data =class_data.merge(subjects[["subject_id","subject_name"]],on="subject_id",how="left")
    class_data =class_data.rename(columns={"subject_name":"subject"})
    class_data =class_data.merge(students[["student_id","name"]],on="student_id",how="left")
```

FIGURE 4.6 A SNIPPET OF LOAD\_APP\_DATA() IN APP\_DATA.PY

The dashboard also provides summary metrics to show the number of students combined, total number currently At Risk, proportion of students At risk and average attendance where available (as shown in figure 4.6.1 and code shown in figure 4.6.2).

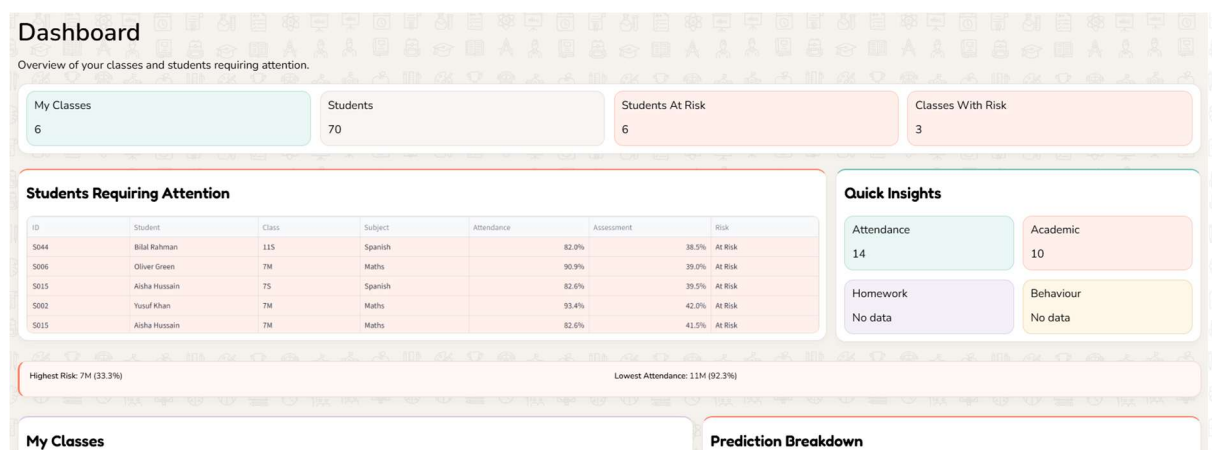


FIGURE 4.6.1 SNAPSHOT OF THE MAIN DASHBOARD

```
#####SUMMARY DATA#####
total_students =dashboard_data["student_id"].nunique()
total_classes =dashboard_data["class_id"].nunique()
#nunique- only count unique students and classes at risk
at_risk_students =dashboard_data[dashboard_data["risk_prediction"]=="At Risk"]["student_id"].nunique()
classes_with_risk =dashboard_data[dashboard_data["risk_prediction"]=="At Risk"]["class_id"].nunique()

#####SUMMARY#####
with st.container(key="dashboard_summary"):
    col1,col2,col3,col4 =st.columns(4)

    with col1:
        with st.container(key="metric_classes"):
            st.metric("My Classes",total_classes)
    with col2:
        with st.container(key="metric_students"):
            st.metric("Students",total_students)
    with col3:
        with st.container(key="metric_risk"):
            st.metric("Students At Risk",at_risk_students)
    with col4:
        with st.container(key="metric_classes_risk"):
            st.metric("Classes With Risk",classes_with_risk)
```

FIGURE 4.6.2 CODE FOR THE SUMMARY METRICS ON DASHBOARD.PY

Areas showing students requiring attention help teachers to identify pupils and follow up on who may need attention without having to manually check every record. A class-level table provides a broader view of students and their current prediction status with assessment and attendance.

The dashboard is supported by visual summaries of risk distribution and attendance, attendance percentage visualisations (as shown in figure 4.6.3 and 4.6.4)

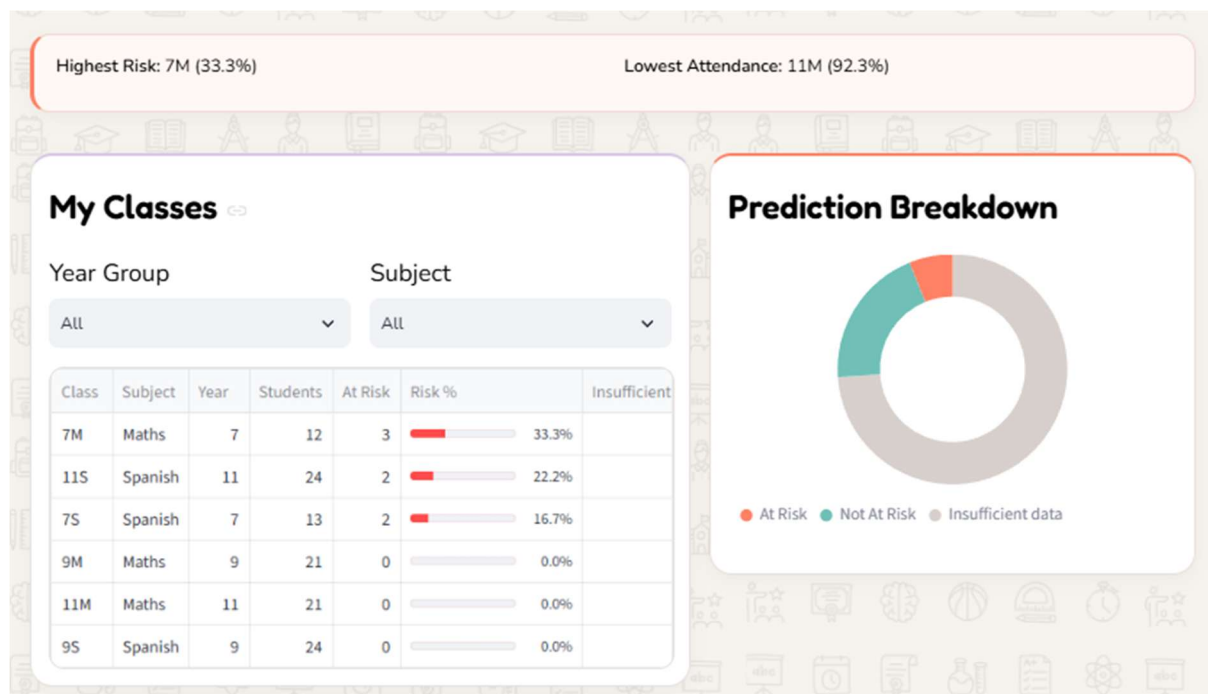


FIGURE 4.6.3 CLASS LEVEL INSIGHTS AND DONUT CHART

```

#add progress column- visually represent percentages as bars
st.dataframe(
    class_table, use_container_width=True, hide_index=True,height=250, column_config={
        "At Risk %":st.column_config.ProgressColumn("Risk %",min_value=0,max_value=100,format="%.1f%%"),
        "Average Attendance %":st.column_config.ProgressColumn("Attendance",min_value=0,max_value=100,format="%.1f%%"),
        "Average Assessment %":st.column_config.ProgressColumn("Assessment",min_value=0,max_value=100,format="%.1f%%")
    }
)

#prediction breakdown chart-round
with st.container(key="risk_chart"):
    st.write("### Prediction Breakdown")

    prediction_data =dashboard_data[["student_id","subject","risk_prediction"]].drop_duplicates()
    prediction_data["risk_prediction"] =prediction_data["risk_prediction"].fillna("Insufficient data")

    prediction_counts =prediction_data["risk_prediction"].value_counts().reset_index()
    prediction_counts.columns =["Risk","Students"]

    #donut chart
    donut =alt.Chart(prediction_counts).mark_arc(innerRadius=60,outerRadius=95).encode(
        #angle for size of each segment
        theta=alt.Theta(field="Students",type="quantitative"),
        #red, green, gray for different risk levels. not risk=green, at risk=red, insufficient data=gray
        color=alt.Color("Risk:N",
            scale=alt.Scale(domain=["At Risk","Not At Risk","Insufficient data"],range=["#FF8266","#70C0B8","#D8D0CC"]),
            legend=alt.Legend(title=None,orient="bottom")
        ),
        tooltip=[alt.Tooltip("Risk:N",title="Prediction"),alt.Tooltip("Students:Q",title="Records")]
    ).properties(height=250)

```

FIGURE 4.6.4 CODE TO HANDLE PROGRESS COLUMN AND DONUT CHART

#### 4.6.2 Class Overview and Student Profiles (170)

The class overview page allows teachers to filter by year, subject or class. The class-level table combines the student data with the model output allowing the students to be presented within a class -colour coordinated with clear risk status.

An individual student can also be selected from the table where the risk category and further insights as to why the prediction was made can be provided: Assessment average, attendance, grade change, previous failure, homework and behaviour. This allows for the prediction to be considered alongside wider contextual information rather than viewing risk scores in isolation (as shown in figure 4.6.5)

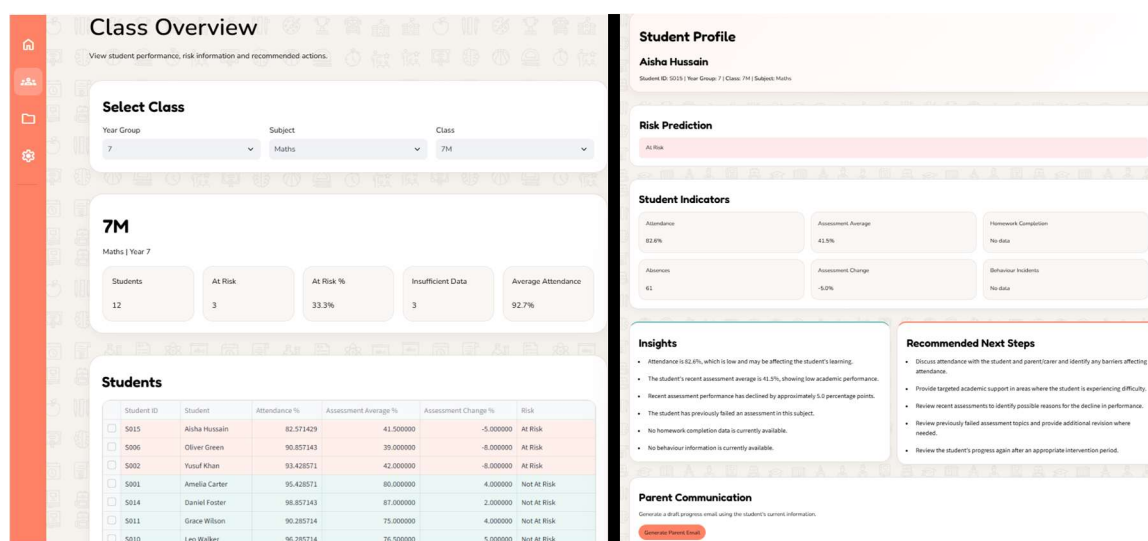


FIGURE 4.6.5 SNIPPET OF THE CLASS OVERVIEW PAGE CONTAINING INSIGHTS, RISK CATEGORIES,

Insights and recommendations are also provided alongside parent/carer email generation, which will be expanded in section 4.7.

## 4.7 Explanation and Recommendation Generation

Although the model predicts a student as At Risk and Not At Risk, it does not itself explain its prediction. Instead, `get_student_insights()` checks for any indicators such as attendance, grade average, grade change, previous failures, homework and behaviour. It then converts them into readable statements (as shown in figure 4.7.1). For example, a low attendance triggers an attendance concern warning.

```
#####STUDENT INSIGHTS#####
#give the information based on student data
def get_student_insights(student):
    insights = []
    #attendance
    attendance = student["attendance_percentage"]

    #if no attendance or if low attendance then insight given
    if pd.isna(attendance):
        insights.append("No attendance data is currently available.")
    elif attendance < 90:
        insights.append(f"Attendance is {attendance:.1f}%, which is low and may be affecting the student's learning.")
    elif attendance < 95:
        insights.append(f"Attendance is {attendance:.1f}% and should continue to be monitored.")
```

FIGURE 4.7.1 A SNIPPET WITHIN `GET_STUDENT_INSIGHTS` THAT HANDLES ATTENDANCE INSIGHTS

Once an insight has been found, `get_recommendations()` will then map those indicators to possible teacher actions, such as “provide targeted support”. If a student falls under the “Insufficient data” category, the system recommend collecting more assessment information before relying on a prediction (as shown in figure 4.7.2).

```
#####RECOMMENDATIONS#####
#recommendations of next steps based on student data
#future improvements will be more personalised to eac student
def get_recommendations(student):
    recommendations = []

    attendance = student["attendance_percentage"]

    ##attendance recommendations
    if not pd.isna(attendance):
        if attendance < 90:
            recommendations.append("Discuss attendance with the student and parent/carer and identify any barriers affecting attendance.")
        elif attendance < 95:
            recommendations.append("Monitor attendance regularly and encourage improved consistency.")
```

FIGURE 4.7.2 A SNIPPET WITHIN `GET_RECOMMENDATIONS` THAT HANDLES ATTENDANCE RECOMMENDATIONS

A parent-email draft edit box is also generated with those insights and recommendations, but the teacher is explicitly informed to review and edit it before use.

## 4.8 Accessibility and Usability Improvements

The final user interface consisted of a consistent general layout, spacing and styling across all pages. Different risk states had a consistent visual highlighting so teachers can scan results quickly (as shown in figure 4.7.3).

Navigation was also simplified with clear page labels and tab filters. Settings allowed users to adjust font size and toggle the OpenDyslexic font (as shown in figure 4.7.4 and 4.7.5). For first-time users, a tutorial slideshow was displayed at the beginning to understand the main workflow of the application (As shown in figure 23). Users who required a quick reminder of the workflow were also able to access the tutorial on the settings page.



FIGURE 4.7.3 LIGHT RED HIGHLIGHTS ON MULTIPLE FEATURES TO INDICATE STUDENT AT

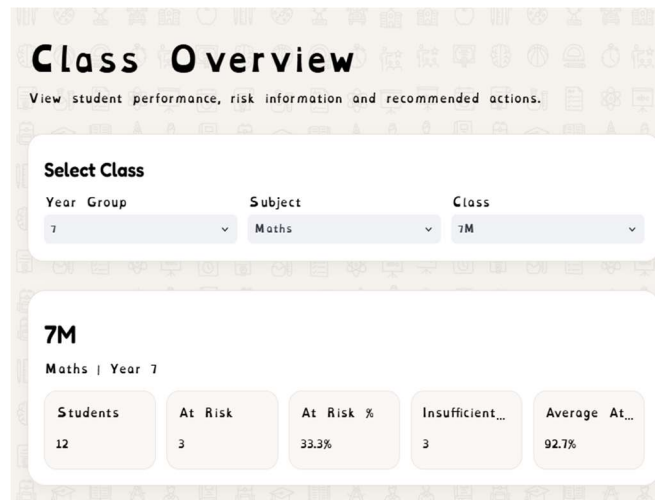


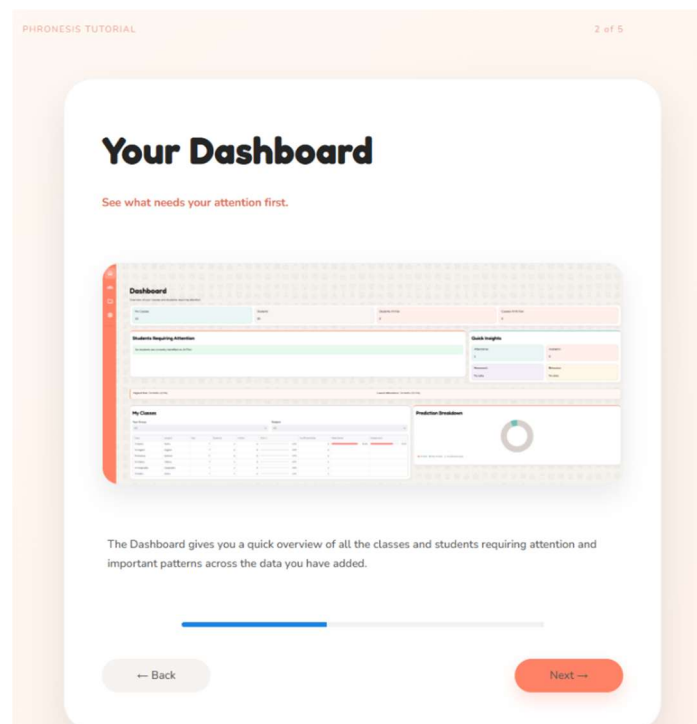
FIGURE 4.7.4 OPENDYSLEXIC FONT ON THE SAME OVERVIEW PAGE. TOGGLE ON SETTINGS PAGE

```
#####FONTS#####
dyslexia_font =st.session_state.get("dyslexia_font",False)
if dyslexia_font:
    font_family ="OpenDyslexic, Arial, sans-serif"
    heading_font ="OpenDyslexic, Arial, sans-serif"
    #get open dyslexic font from CDN
    #fix!- bold and not bold
    st.markdown("""<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/@fontsource/opendyslexic@5.3.0/400.css">
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/@fontsource/opendyslexic@5.3.0/700.css">
    """, unsafe_allow_html=True)
```

FIGURE 4.7.5 CODE SNIPPET OF THE IMPLEMENTATION OF THE OPEN DYSLEXIC FONT THROUGH A CONTENT DELIVERY NETWORK (CDN). THIS ALLOWS FONT FILE TO BE LOADED FROM AN EXTERNAL SERVER RATHER THAN BEING STORED INSIDE PROJECT FILES.

## 4.9 Final Implementation Outcome

Overall, the final system progressed from the traditional rule-based prototype to a complete teacher-used Streamlit web application. The final version contains student and class data management, assessment/attendance processing, Random Forest risk prediction, and clear outcomes identifying students as At Risk, Not At Risk or Insufficient data. A comprehensive dashboard was also included with class and student views, rule-based insights and recommendations, an editable parent email draft and accessibility features.



**FIGURE 4.7.6 THE TUTORIAL POP-UP THAT IS PROVIDED WHEN A NEW USER SIGNS IN FOR THE FIRST TIME**

Therefore, this final implementation supports a full user workflow as originally intended for Phronesis. Where data is entered by a teacher, then validated and processed, transformed into features for the model, predicted with a risk category and then presented alongside relevant evidence and suggested next steps. This moves the project from simply training a prediction model, but provides a more complete decision support process for teachers.

## Chapter 5: Evaluation (2477 words)

### 5.1 Evaluation Overview and Approach

The evaluation of Phronesis, as a whole, was a mixed approach rather than relying on a single method. It was divide into two main areas: Technical evaluation and user evaluation.

Technical Evaluation referred to the ML performance testing, dataset experiment testing, functional and integration testing, incomplete-data testing and edge-case testing. The information derived from these tests

were used to assess whether the system behaved correctly and whether the model was providing reliable predictions.

User Evaluation included a task-based questionnaire completed by fourteen participants, thirteen with consent for data to be used and one without consent given and thus excluded from analysis. Three follow up interviews were also carried out to provide deeper qualitative feedback.

The quantitative results and the questionnaires gave teacher first insight into this application and responses allowed the model performance to be assessed. Interviews, however, provided a deeper explanation of what was preferred in this application and what was unclear or what should be improved.

## 5.2 Machine Learning Evaluation

The ML component was evaluated with more metrics than just accuracy due to the unbalanced dataset. Only 15% of the dataset was labelled as At Risk. As a result, F1, ROC-AUC, recall, precision, Average Precision and confusion matrices were used in multiple experiments. This conclusion came about as the majority class baseline achieved an 84.62% accuracy but a 0% recall and F1 score. This was because it simply predicted every student as Not At Risk.

The cross-validation results of the Tuned Random Forest showed the following results. Accuracy: 94.22%, Precision: 77.78%, Recall: 87.50%, F1: 82.35%, ROC-AUC: 0.9665, Average Precision: 0.8584. A confusion matrix was also drawn to show 419 true negatives, 20 false positives, 10 false negatives and 70 true positives. That means it correctly detected 70 out of the 80 possible At-Risk students and only produced 20 unnecessary alerts.

Different variations of models were also compared. The Balanced Random Forest model achieved a higher recall (92.5%) but a lower precision (69.16%) and a lower F1 score (79.14%). The tuning of the random forest model reduced the false positives from 33 to 20 but increased false negatives from 6 to 10. It was then identified that tuning improved balance overall. The Depth-2 Decision Tree provided very close results to the tuned random forest with an F1 score of 82.08% recall 88.75%. This shows that the random forest model was not dramatically out-performing performing the other models.

The final holdout test provided the following results:

Accuracy: 88.46%, Precision: 58.06%, Recall: 90%, F1: 70.59%, ROC-AUC: approximately 0.932, Average Precision: approximately 0.852, 18/20 At Risk students were detected, 2 At Risk students were missed. 13 false positives occurred (as shown in figure 26). The results were also shown in a confusion matrix (as shown in Figure 27)

These outcomes proved to be useful as, although the final holdout shows the model performs well, they also showed a loss in performance when tested on unseen data. 13 false positives occurred, though it was established that they were less impactful than if an At Risk student was missed.

Recall is an important metric as, if an At Risk student is incorrectly classified as Not At Risk, it could mean that some students who actually do need support are overlooked. However, if recall was maximised, it would create too many false alerts and increase the teacher workload, defeating the purpose of Phronesis.

The final model therefore represented a compromise so that it detects At Risk pupils and limits unnecessary intervention. The models Brier score was approximately 0.0639, but some middle-range probabilities overestimated the observed risk. Because of this, Phronesis does not show the raw model as an exact “probability of failure”, it presents At Risk / Not At Risk alongside contextual evidence instead.

| Model | Accuracy                      | Precision          | Recall             | F1     | ROC-AUC            | Average Precision  | Predicted At Risk  |    |
|-------|-------------------------------|--------------------|--------------------|--------|--------------------|--------------------|--------------------|----|
| 1     | Logistic Regression 2B @ 0.40 | 0.9364161849710982 | 0.8051948051948052 | 0.775  | 0.7898089171974523 | 0.9729498861047836 | 0.8694382490204141 | 77 |
| 2     | Decision Tree 2B depth 2      | 0.9402697495183044 | 0.7634408602150538 | 0.8875 | 0.8208092485549133 | 0.9615888382687927 | 0.8138715328107275 | 93 |
| 3     | Tuned Random Forest 2B        | 0.9421965317919075 | 0.7777777777777778 | 0.875  | 0.8235294117647058 | 0.9665148063781321 | 0.8584478519181583 | 90 |

FIGURE 5.2 FINAL MODEL COPARISON TABLE

```

print("Final Random Forest confusion matrix:")
print(final_test_confusion)

print("Predicted At Risk:",final_test_predicted_at_risk)
print("Accuracy:",final_test_accuracy)
print("Precision:",final_test_precision)
print("Recall:",final_test_recall)
print("F1:",final_test_f1)
print("ROC-AUC:",final_test_roc_auc)
print("Average Precision:",final_test_a_precision)

```

```

Final Random Forest confusion matrix:
[[97 13]
 [ 2 18]]
Predicted At Risk: 31
Accuracy: 0.8846153846153846
Precision: 0.5806451612903226
Recall: 0.9
F1: 0.7058823529411765
ROC-AUC: 0.9322727272727271
Average Precision: 0.8515800416649313

```

FIGURE 5.2.1 FINAL CONFUSION MATRIX

## 5.3 Functional and Integration Testing

Functional testing was carried out to check whether each main feature behaved correctly. Integration testing checked whether the data could move throughout the full pipeline:

Upload/input -> validation -> feature engineering -> merge -> prediction -> dashboard/profile output.

The strongest cases from the testing include:

Duplicate class names with inconsistent whitespace/capitalising. For example “7B Maths” and “7B Maths” would both have bypassed the duplicate detection (as shown in the figures below). The fix normalised the whitespaces before comparisons were made.

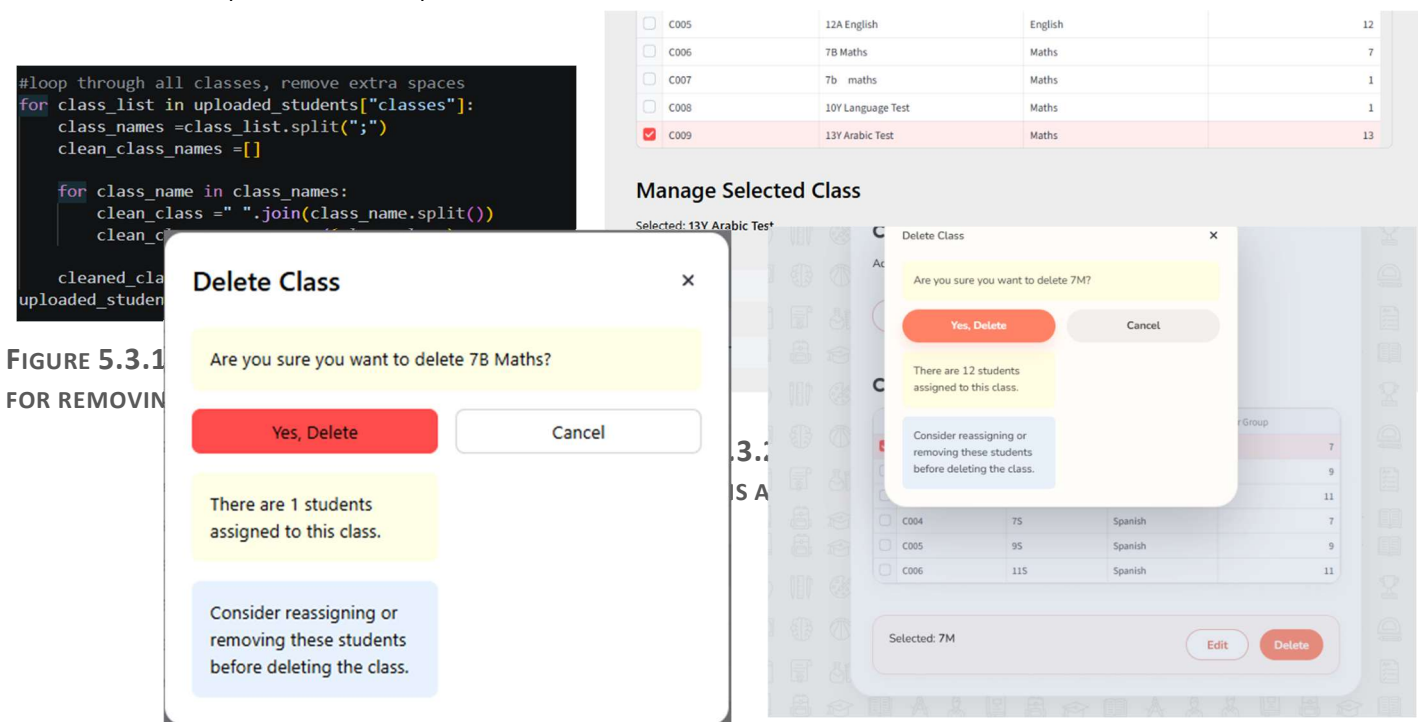


FIGURE 5.3.1  
FOR REMOVING

FIGURE 5.3.3 SCREENSHOT OF THE ERROR MESSAGE WHEN UNSAFE CLASS DETECTION WAS FIXED.

FIGURE 5.3.4 FINAL APPLICATION SCREENSHOT CONTAINING THE SAME ERROR HANDLING

Another case was regarding a class that could be deleted while students were still linked to it. The fix blocked the deletion while class\_students still referenced the class, making it no longer possible in the final version. The figures below demonstrate the fix by showing an error message of how many students are currently enrolled within that class.

Another issue found was that validation originally did not prevent dates in the future. A rule was added so assessment dates must be on or before the current date. Assessment validation was originally duplicated inside both validate\_assessment\_data() and process\_assessments(). I changed process\_assessments() to call the validation function instead. This reduced repeated code and means any future validation changes only need to

- **Assessment validation test summary:** 30 validation tests were carried out. On the first run, 29 tests passed and 1 failed because future assessment dates were still being accepted. A future-date validation check was added, and the failed test was repeated. The retest passed, meaning all 30 assessment validation tests now behave as expected.

FIGURE 5.3.5 THE FIRST TEST BATCH WITH FIXES

be made in one place. Existing tests were rerun afterwards to check that the behaviour had not changed. (as documented below)

Attendance validation was tested against invalid IDs, missing values, non-numeric or non-whole-number sessions values, cases where sessions\_attended > total\_sessions and duplicate attendance records. These were

| ATTENDANCE VALIDATION TESTS: |                                      |                |              |             |  |  |
|------------------------------|--------------------------------------|----------------|--------------|-------------|--|--|
|                              | test_case                            | expected_valid | valid_record | test_passed |  |  |
| 0                            | normal valid attendance              | True           | True         | True        |  |  |
| 1                            | all sessions attended                | True           | True         | True        |  |  |
| 2                            | zero sessions attended               | True           | True         | True        |  |  |
| 3                            | sessions attended greater than total | False          | False        | True        |  |  |
| 4                            | negative sessions attended           | False          | False        | True        |  |  |
| 5                            | zero total sessions                  | False          | False        | True        |  |  |
| 6                            | negative total sessions              | False          | False        | True        |  |  |
| 7                            | non numeric total sessions           | False          | False        | True        |  |  |
| 8                            | non numeric sessions attended        | False          | False        | True        |  |  |
| 9                            | missing total sessions               | False          | False        | True        |  |  |
| 10                           | missing sessions attended            | False          | False        | True        |  |  |
| 11                           | missing student id                   | False          | False        | True        |  |  |
| 12                           | both session values missing          | False          | False        | True        |  |  |
| 13                           | null total sessions                  | False          | False        | True        |  |  |
| 14                           | null sessions attended               | False          | False        | True        |  |  |

then tested to stop the impossible attendance data entering the prediction pipeline. The terminal output below demonstrates the attendance validation tests that were carried out.

**FIGURE 5.3.6 A** TERMINAL OUTPUT OF THE ATTENDANCE VALIDATION TESTS ALONGSIDE THEIR OUTPUTS

Another case was during prediction integration. Initially complete students received predictions and incomplete students also remained visible. The incomplete students were not passed into the Random Forest. multiple-subject students were handled separately. The integration scenarios all passed.

Testing revealed several faults that would not have appeared when using only the pre-populated dataset. Fixing these issues improved the overall robustness of the project as well as the user experience. However, testing was limited in a sense that it was not completed on real school data in a live school environment.

## 5.4 Edge-Case and Incomplete-Data Testing

The purpose of Edge-case testing was to test how Phronesis behaved when student information was incomplete. The demo teacher dataset provided in the application deliberately included missing attendance, insufficient assessment history, students with no assessment records, classes with attendance but no assessment data, students enrolled in multiple subjects. Students with missing required model features should have remained visible but receive Insufficient data rather than being forced into At Risk or Not At Risk. This tested that missing values were not replaced by arbitrary zeros. Testing with multiple subjects checked that assessment trends remained subject-specific rather than crossing over. The outcome showed that the system handled incomplete data without crashing and would avoid generating unreliable predictions. However, this was still based on fictional demonstration data rather than a genuine incomplete record from a live school.

This testing showed that the system could handle incomplete and unusual inputs without forcing a prediction, though further practice is required with real school data sets before handled in practice reliably.

## 5.5 User Evaluation Method

The purpose of user evaluation was to check the ease of use and usefulness from the perspective of potential users (teachers). 14 Participants were shown the prototype and were asked to complete a short set of tasks before answering the questionnaire. The tasks included are shown in the image from the questionnaire below.

Before answering the following questions, please spend a few minutes exploring Phronesis using the demonstration account provided. Please try to complete the following tasks:

1. View the Dashboard and identify any classes or students requiring attention.
2. Open the Class Overview page and select a class.
3. Select an At Risk student and review the indicators, insights and recommended next steps.
4. Select a student with Insufficient data and review how the application handles missing information.
5. Review a student who is Not At Risk.
6. Generate a draft parent/carers communication for a student.
7. Explore the Data Management area to see how student, class, assessment and attendance data can be managed.



FIGURE 5.4.1 SNIPPET OF THE QUESTIONNAIRE INSTRUCTIONS PROVIDED.

Fourteen participants completed the questionnaire, though thirteen consented to their anonymised responses being used in the report. Questions used a five-point Likert scale from Strongly Disagree to Strongly Agree, and covered areas like navigation, the clarity of risk categories, usefulness of explanations and recommendations and the overall usability within a school environment. The final 3 questions were open-ended to gather a range of quantitative and qualitative analysis. An example of an open-ended question is provided below.

Q6 10. Which feature did you find most useful, and why? Essay

---

Answers

The student risk section because it brought everything together in one place.

Probably the attendance and assessment info together. It made patterns easier to notice.

I liked the fact it showed when there wasn't enough data instead of pretending the result was certain

The colour coding helped me scan the class list faster

I found the data management section useful because everything was in one place and quite easy to edit.

The parent email was good because it provides a good starter

I liked the class overview because I could compare students really quickly

The dashboard was good because it was quick for me to see which classes needed looking at

The student risk section because it brought everything together in one place.

The student indicators were good because I didn't have to open lots of different pages.

The insufficient data message was actually useful because it showed the system wasn't just making a guess.

The student indicators to see which ones are at risk, not risk etc.

Everything is centralised: the homework, the date of each student, attendance and behaviour. This saves time from going on separate spreadsheets for each individual thing.

I loved the parent email generator, will save me so much time going back and forth looking at data

FIGURE 5.4.2 EXAMPLE OF AN OPEN-ENDED ESSAY QUESTION IN THE

Three follow-up interviews were also used to explore participants opinions in further detail. I also carried out born accessible design and elements of radical inclusion with the accessibility being implemented during development rather than being added only at the end. Teachers with dyslexia were consulted directly about the most readable and suitable font choices and font sizes for this application. This allowed accessibility decisions to be informed by users with lived experience rather than relying only on assumptions and search engine research.

However, the limitation of this user evaluation was the small sample size. 13 participants opinions are most definitely not representative of all teachers. This evaluation was suitable for the short-term prototype testing rather than for long-term use in a real school environment. Hence, a stronger evaluation would require a larger and more diverse sample and with users with a wider range of accessibility needs.

## 5.6 User Evaluation Results

Thirteen questionnaire responses were included in the final analysis. Overall, the results were used to assess the usefulness from a teachers perspective. All respondents were able to complete the main testing tasks. Below are the main quantitative findings also shown in the figure.

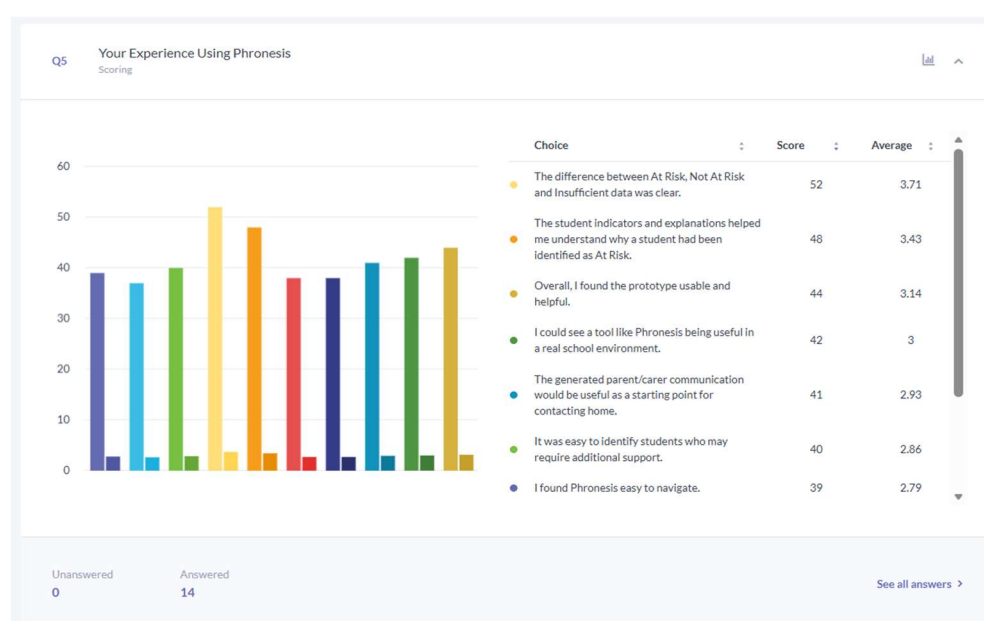


FIGURE 5.6.1 (1-5) LIKERT SCALE FINDINGS

The strongest areas were the clarity of At Risk/ Not At Risk/ Insufficient data with an average score of 3.71. Student indicators and explanations with a score of 3.43. The relatively weaker areas were dashboard clarity, with a score of 2.64, recommended next steps with a score of 2.71.

This method was very useful as it showed specific strengths and weaknesses. Throughout the questionnaire, the positive themes were summarised. One respondent specifically mentioned that having homework, attendance and behaviour centralised would save time compared with checking separate spreadsheets. Another respondent particularly liked the Insufficient Data feature as the system did not “Pretend” to know a

result. Others preferred the parent email generator. In an interview, a teacher mentioned the colour coding allowed for fast scanning of data which saved time.

However, there were some areas of confusion highlighted. Some respondents wanted more clarity of the meaning of At Risk, or whether assessment percentages used all results or only recent assessments (as shown in the figure below). Although, these are already predefined in the planning, users also sought that information as to how the model is coming to its conclusion.

The screenshot shows a survey interface. At the top, there is a question number 'Q7' and the text '11. Was anything confusing, difficult to use, or unclear?' followed by the word 'Essay'. Below this is a section titled 'Answers' which contains a list of user responses. The responses are as follows:

- The difference between an insight and a recommendation could maybe be clearer.
- The upload section was clear once I saw the templates but I probably wouldnt know the csv format without them.
- Nothing major, just some wording could be shorter.
- Class filters were a bit confusing, but got there in the end
- Maybe explain what at risk actually means in terms of the model.
- Some wording could be shorter, nothing major
- There was quite a lot on the student page so it was a bit confusing at first, but it was fine after thay
- No
- I wasnt sure if the percentages were based on all assessments or just the most recent ones.
- Nope
- The dashboard had a lot of information so maybe the most important bits could stand out even more.
- I found it easy to navigate as an assistant teacher
- If you aee using a phone then everything can be cluttered.
- I feel like tbales for assessments could be clearer when analysing it

**FIGURE 5.6.2 INDICATES USER RESPONSES AS TO WHAT WAS CONFUSING OR UNCLEAR**

Overall, these results suggest that core concepts of combining student information and highlighting risk were understood. the lower scores for dashboard clarity and recommendations, together with the written comments, indicate that some information needs to be presented abit more clearly. Future development should focus on improving the explanation of the model outputs and expressing information of model outputs in an easier way.

## 5.7 Evaluation Against Project Aims and User Requirements

The overall aim of Phronesis was to bridge educational data and teacher decision making by identifying students falling under academic risk, explaining contributing factors and suggesting interventions.

The final system achieved the main MUST HAVE requirements: Teacher login, dashboard with summary statistics, individual student and class view, CSV and manual data input, attendance and assessment analysis, At Risk/ Not At Risk identification, explanations and recommendations based on each prediction type and a parent email generator.

This is evidently supported by the ML evaluation indicating the final model detected 18/20 At Risk students on holdout data. Functional and integration tests showed a complete data workflow with complete and incomplete data being handled without forced/invalid predictions. Questionnaire feedback showed that users understood the main risk states and valued centralised data and student indicators.

However, dashboard clarity did score lower than other areas. Users still wanted clearer explanations of what At Risk means and how the outputs are produce. Although an out-of-scope requirement, no MIS integration of

intervention tracking was implemented. This further reinforced one of the core aims of not replacing an MIS system.

Overall Phronesis met the central project aims and majority of the high-priority user requirements. However, these aims were tested at a prototype level and not validated on a school system. Evaluation also showed that improved dashboard presentation and clearer communication would make a significant improvement.

## 5.8 Successes and Strengths

The strongest success was replacing the original rule-based prototype into a complete end-to-end ML prediction and support system. Phronesis successfully connects the teacher-entered data to feature engineering, uses a Random Forest prediction model, provides explanations and recommendations.

The model achieved a strong recall on the final holdout test, which demonstrates its usability on unseen data. Incomplete records were also handled safely through the “Insufficient data category” which prevented forced and unreliable predictions. Subject specific processing allowed for multiple subjects and insight into how student is performing within that subject only. All information was centralised with accessibility features were also considered which were also approved by users in evaluations.

Overall, the main strength of Phronesis was that it is not just an application making a prediction, but it combines context and possible action within one teacher workflow.

## 5.9 Limitations and Failures

The model was trained on the UCI Portuguese student dataset, and so it may not generalise well to UK school contexts and environments. Further testing is required within a real UK school data. The final model relied heavily on recent assessment performance; this means it may identify underperformance more than predicting the risk as early as possible. The hold out test showed a drop in F1 scores from the cross-validation result which caused a loss in generalisation and also there still remained 13 false positives (students incorrectly flagged as At Risk). The recommendation system was rule-based rather than model generated. Its suggested interventions also were not validated with long term classroom outcomes and so Phronesis has not completely proven to give the most appropriate recommendations.

In terms of evaluation, user evaluation only involved 13 consenting participants and so findings cannot represent all teachers. Testing only used fictional/demo data rather than live school data and so currently not suitable for a production school environment. The current CSV-based storage and authentication in terms of security are also not valid for a school environment. Furthermore, accessibility testing was limited to a small number of users and was not formally tested against full accessibility standards.

These limitations suggest Phronesis should be used as a proof-of-concept rather than implementing it directly into a school system. The project clearly demonstrates its feasibility, but further validation, testing and infrastructure is required for it to be practically used with sensitive school data.

## 5.10 Future Improvements and Extensions

Future development would be to focus on Phronesis being more school environment ready by validating on a larger, UK representative school dataset. Improvements to the front-end including dashboard design and user communication would be given. Further extensions of this project could be using a ML model for predictions and recommendations, complete accessibility testings and improved security infrastructure and possible MIS integration ready for school deployment.

## 5.11 Evaluation Summary

Overall, evaluation provided useful insights into how Phronesis met its main project aims, while also identifying limitations in generalisation, interface clarity and real-world deployment, in order to guide future development.

# Chapter 6: Conclusion (601 words)

## 6.1 Project Outcome

The original problem that influenced the development of this app was that teachers have student information fragmented and spread across multiple sources making comprehensive interpretation difficult and time-consuming. Hence Phronesis was developed as a teacher-facing decision support application.

The initial prototype was a simple rule-based support system which was then developed substantially to contain a complete workflow:

Upload/input data -> validation -> feature engineering -> merge -> ML prediction -> contextual information -> suggested teacher action.

This allowed data entered by the teacher to be transformed into exact model features required for prediction and then presented with supporting context and possible actions. Student predictions were presented into 3 categories: At Risk, Not At Risk and Insufficient data.

## 6.2 What the Project Demonstrated

This project demonstrated a relatively simple educational dataset (attendance and assessment record) being transformed into a complete teacher decision support workflow. Feature engineering allowed the raw assessment and attendance records to be converted into model-ready inputs, without having the teacher input/explain anything else other than the records. A trained ML model could identify many At Risk students. However, careful consideration on interpretation is required because false positives and false negatives remained. Predictions combined with contextual indicators and recommendations made the output more useful than the classification alone, though teachers are also warned that this is a decision support application and not a decision replacement. The Insufficient data category reduced the risk of unreliable predictions being produced from incomplete records. Finally, user feedback showed that centralisation, colour-coding, risk category clarity were all useful to improve user experience

To conclude, phronesis demonstrated the ability to combine multiple sources of educational data to ML mode and decision support all in one simple workflow.

## 6.3 Critical Reflection and Further Work

The strongest limitation is that the model was trained on UCI Portuguese data and so not yet validated for a UK based schooling system. And so future work would be to train on a larger UK school dataset. Evaluation should also be completed by a larger and more diverse range of educational professionals.

The model also relied heavily on recent assessment performance, and so it may be identifying current underperformance more than very early academic risk, additional indicators could be explored in future models. The reduction in F1 performance on the final holdout set also shows that further validation is required before deployment.

As the recommendations are rule-based, further research will need to be conducted in order to validate this against long term classroom outcomes. Also, the current CSV storage and authentication is more prototype level and so stronger authentication, secure storage, encryption and a production database would be required for future work. Further work would be to implement this into an existing MIS as an extension.

## 6.4 Final Conclusion

To conclude, Phronesis was a challenging but valuable project, particularly because models had to be trained, compared, refined and then integrated into a working application. Experimenting with feature sets, class imbalance, model tuning and integration developed my understanding of the full data-science lifecycle.

Given more time, the front-end and overall functionality could have been developed further. However, the ML model training and research was thoroughly completed.

As a teacher, I can see practical value in the system, particularly because I have experienced the difficulty of interpreting fragmented student data. The concept is also currently being considered within my school for possible future use or trial, although further validation would be required before actual deployment.

This project showed me that what began as an academic project ultimately developed into a concept with potential value beyond the coursework itself.

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## APPENDIX A

### Appendix A.1 Functional Requirements

| ID  | Requirement         | Description                                                                                   | Research Justification                                                                | Source                             | Priority |
|-----|---------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------|----------|
| FR1 | Dashboard Homepage  | Display key alerts, student risk summaries and overall class insights.                        | Teachers require quick access to key information without navigating multiple screens. | Dashboard Research; Survey Q8, Q14 | Must     |
| FR2 | Student List        | Display all students within a selected class.                                                 | Supports rapid identification of students requiring additional support.               | Survey Q7, Q8                      | Must     |
| FR3 | Student Profile     | Display attendance, assessment, behaviour and homework information for an individual student. | Teachers reported that information is often spread across multiple systems.           | Interviews; Survey Q8              | Must     |
| FR4 | Attendance Analysis | Analyse attendance data and identify declining attendance patterns.                           | Attendance is a recognised indicator of academic risk.                                | Attendance Literature              | Must     |
| FR5 | Assessment Analysis | Analyse assessment performance and identify declining attainment.                             | Assessment data was the most frequently used data source.                             | Survey Q4                          | Must     |
| FR6 | Behaviour Analysis  | Analyse behaviour incidents and identify repeated concerns.                                   | Behaviour records are regularly used to monitor student wellbeing and progress.       | Survey Q4                          | Should   |
| FR7 | Homework Analysis   | Analyse homework completion and identify missing work patterns.                               | Homework completion was identified as an important performance indicator.             | Survey Q4                          | Should   |
| FR8 | At-Risk Alerts      | Automatically identify students requiring additional support.                                 | At-risk alerts were the most requested dashboard feature.                             | Survey Q11                         | Must     |
| FR9 | Risk Score          | Calculate and assign Low, Medium or High risk categories.                                     | Supported by Educational Data Mining and student performance prediction research.     | EDM Literature                     | Must     |

|      |                              |                                                                                         |                                                                          |                                |        |
|------|------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------|--------|
| FR10 | Explainability Engine        | Explain why a student has been identified as at risk using plain-language explanations. | Teachers requested explanations rather than unexplained predictions.     | Survey Q8, Q14; XAI Literature | Must   |
| FR11 | Root Cause Analysis          | Identify likely factors contributing to student underperformance.                       | Teachers reported difficulty understanding why students were struggling. | Interviews; Survey Q8          | Must   |
| FR12 | Student Summary              | Generate concise summaries of student performance and identified concerns.              | Teachers requested quicker interpretation of student information.        | Survey; Interviews             | Must   |
| FR13 | Intervention Recommendations | Recommend evidence-based interventions tailored to identified risks.                    | Teachers reported uncertainty regarding appropriate next steps.          | Survey Q8; EEF Research        | Must   |
| FR14 | Parent Email Generator       | Generate draft parent communication based on identified concerns.                       | Teachers requested improved parent communication tools.                  | Survey Q15                     | Should |
| FR15 | Dashboard Visualisations     | Display charts showing student trends and overall class performance.                    | Visualisation supports quicker interpretation of educational data.       | Dashboard Literature           | Should |
| FR16 | Search and Filter            | Search students and filter by class or risk category.                                   | Reduces time spent locating individual students.                         | Teacher Workload Research      | Should |
| FR17 | Intervention Tracking        | Record completed interventions and monitor outcomes over time.                          | Encourages evidence-based follow-up rather than one-off interventions.   | Intervention Literature        | Could  |
| FR18 | Export Student Report        | Export student summaries for meetings or record keeping.                                | Supports reporting and communication with staff and parents.             | Interviews                     | Could  |

## Appendix A.2 Data Management Requirements

| ID   | Requirement             | Description                                                       | Research Justification                                                | Priority |
|------|-------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------|----------|
| FR19 | Class-Based Data Upload | Upload datasets for a selected teaching class.                    | Teachers commonly organise tracking information by class.             | Must     |
| FR20 | Assessment Upload       | Upload assessment data using a standard CSV template.             | Enables consistent attainment analysis.                               | Must     |
| FR21 | Attendance Upload       | Upload attendance records using a standard CSV template.          | Attendance is a key indicator within the risk model.                  | Must     |
| FR22 | Behaviour Upload        | Upload behaviour records using a standard CSV template.           | Behaviour contributes to student risk analysis.                       | Should   |
| FR23 | Homework Upload         | Upload homework completion records using a standard CSV template. | Homework completion was identified as a useful performance indicator. | Should   |
| FR24 | Download Templates      | Download standardised CSV templates before uploading data.        | Reduces formatting errors during upload.                              | Must     |
| FR25 | Data Validation         | Validate uploaded files for missing columns and invalid values.   | Improves reliability and reduces processing errors.                   | Must     |
| FR26 | Manual Record Editing   | Allow teachers to amend individual records after upload.          | School records may require correction after import.                   | Should   |

|      |                              |                                                                     |                                                                     |        |
|------|------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------|--------|
| FR27 | Upload New Assessment Period | Add assessment data from future assessment windows.                 | Supports longitudinal tracking of student progress.                 | Must   |
| FR28 | Replace Existing Dataset     | Replace previously uploaded datasets when corrections are required. | Educational data is updated regularly throughout the academic year. | Should |

### Appendix A.3 Future Requirements (Out of Scope)

| ID   | Requirement                       | Justification                                                                  | Priority |
|------|-----------------------------------|--------------------------------------------------------------------------------|----------|
| FR29 | SIMS Integration                  | Requires external APIs and school-level permissions beyond the project scope.  | Won't    |
| FR30 | Arbor Integration                 | Outside the available development timescale.                                   | Won't    |
| FR31 | BromCom Integration               | Outside the available development timescale.                                   | Won't    |
| FR32 | Live Parent Messaging             | Raises safeguarding, authentication and privacy considerations.                | Won't    |
| FR33 | Machine Learning Prediction Model | Requires significantly larger real-world datasets for training and validation. | Won't    |
| FR34 | Multi-School Deployment           | Beyond the scope of the proposed prototype.                                    | Won't    |

### Appendix A.4 Requirements Traceability Matrix

| Requirement ID | Requirement                  | Related User Story | Design Section               | Implementation                |
|----------------|------------------------------|--------------------|------------------------------|-------------------------------|
| FR1            | Dashboard Homepage           | US2                | 3.8 User Interface Design    | Streamlit dashboard homepage  |
| FR2            | Student List                 | US2                | 3.8 User Interface Design    | Student selection page        |
| FR3            | Student Profile              | US2                | 3.8.2 Student Profile Design | Student profile screen        |
| FR4            | Attendance Analysis          | US1                | 3.4 Data Design              | Attendance dataset analysis   |
| FR5            | Assessment Analysis          | US1                | 3.4 Data Design              | Assessment dataset analysis   |
| FR6            | Behaviour Analysis           | US1                | 3.4 Data Design              | Behaviour dataset analysis    |
| FR7            | Homework Analysis            | US1                | 3.4 Data Design              | Homework dataset analysis     |
| FR8            | At-Risk Alerts               | US2                | 3.5 Risk Analysis Engine     | Automatic risk identification |
| FR9            | Risk Score                   | US2                | 3.5 Risk Analysis Engine     | Risk scoring algorithm        |
| FR10           | Explainability Engine        | US3                | 3.6 Explainability Engine    | Plain-language explanations   |
| FR11           | Root Cause Analysis          | US3                | 3.6 Explainability Engine    | Rule-based explanation logic  |
| FR12           | Student Summary              | US2                | 3.8.2 Student Profile Design | Student summary panel         |
| FR13           | Intervention Recommendations | US4                | 3.7 Recommendation Engine    | Recommendation engine         |

|      |                              |     |                           |                                   |
|------|------------------------------|-----|---------------------------|-----------------------------------|
| FR14 | Parent Email Generator       | US4 | 3.7 Recommendation Engine | Parent email template generation  |
| FR15 | Dashboard Visualisations     | US2 | 3.8.1 Dashboard Design    | Dashboard charts and summaries    |
| FR16 | Search and Filter            | US2 | 3.8 User Interface Design | Student search and filtering      |
| FR17 | Intervention Tracking        | US4 | Future Development        | Not implemented (prototype scope) |
| FR18 | Export Student Report        | US5 | Future Development        | Not implemented (prototype scope) |
| FR19 | Class-Based Data Upload      | US1 | 3.4 Data Design           | CSV upload process                |
| FR20 | Assessment Upload            | US1 | 3.4 Data Design           | Assessment CSV import             |
| FR21 | Attendance Upload            | US1 | 3.4 Data Design           | Attendance CSV import             |
| FR22 | Behaviour Upload             | US1 | 3.4 Data Design           | Behaviour CSV import              |
| FR23 | Homework Upload              | US1 | 3.4 Data Design           | Homework CSV import               |
| FR24 | Download Templates           | US1 | 3.4 Data Design           | Standard CSV templates            |
| FR25 | Data Validation              | US1 | 3.4 Data Design           | Upload validation checks          |
| FR26 | Manual Record Editing        | US1 | Future Development        | Not implemented (prototype scope) |
| FR27 | Upload New Assessment Period | US1 | 3.4 Data Design           | Multiple assessment uploads       |
| FR28 | Replace Existing Dataset     | US1 | 3.4 Data Design           | Dataset replacement function      |

## APPENDIX B

### Appendix B.1 Teacher Survey Questions:

#### What is your current role?

- ☐ Classroom Teacher
- ☐ Head of Department
- ☐ Senior Leader
- ☐ Teaching Assistant
- ☐ Other

#### How many years have you worked in education?

- ☐ 0–2 years
- ☐ 3–5 years
- ☐ 6–10 years
- ☐ 10+ years

#### Which phase do you teach?

- ☐ Primary
- ☐ Secondary
- ☐ College/Sixth Form
- ☐ Other

**Which student data do you regularly use? (Select all that apply.)**

- ☐ Attendance
- ☐ Assessment Results
- ☐ Behaviour Records
- ☐ Homework Completion
- ☐ SEND Information
- ☐ Safeguarding Information
- ☐ Other

**How often do you analyse student data?**

- ☐ Daily
- ☐ Weekly
- ☐ Monthly
- ☐ Termly
- ☐ Rarely

**Which systems do you currently use?**

- ☐ SIMS
- ☐ Bromcom
- ☐ Arbor
- ☐ Google Sheets
- ☐ Other

**How easy is it to identify students who need support using your current systems?**

- ☐ **Scale:**
- ☐ Very Easy
- ☐ Easy
- ☐ Neutral
- ☐ Difficult
- ☐ Very Difficult

**What are your biggest challenges when using student data to inform teaching and decision-making?**

- ☐ Free-text response

**"School data is collected but does not always lead to meaningful action."**

**To what extent do you agree?**

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree

- ☐ Strongly Disagree

**How much time do you spend each week reviewing student data?**

- ☐ Less than 1 hour
- ☐ 1–3 hours
- ☐ 4–6 hours
- ☐ 7–10 hours
- ☐ More than 10 hours

**Which features would be most useful in a student analytics dashboard? (Select up to 5.)**

- ☐ At-risk student alerts
- ☐ Attendance trend analysis
- ☐ Behaviour trend analysis
- ☐ Performance predictions
- ☐ Intervention recommendations
- ☐ Parent email generation
- ☐ Student summaries
- ☐ Automated reports
- ☐ AI explanations of performance changes
- ☐ Intervention tracking

**How useful would AI-generated recommendations be for your role?**

- ☐ Very Useful
- ☐ Useful
- ☐ Neutral
- ☐ Not Useful
- ☐ Not Useful At All

**Would you trust an AI recommendation if the system clearly explained how it reached the recommendation?**

- ☐ Yes
- ☐ Maybe
- ☐ No

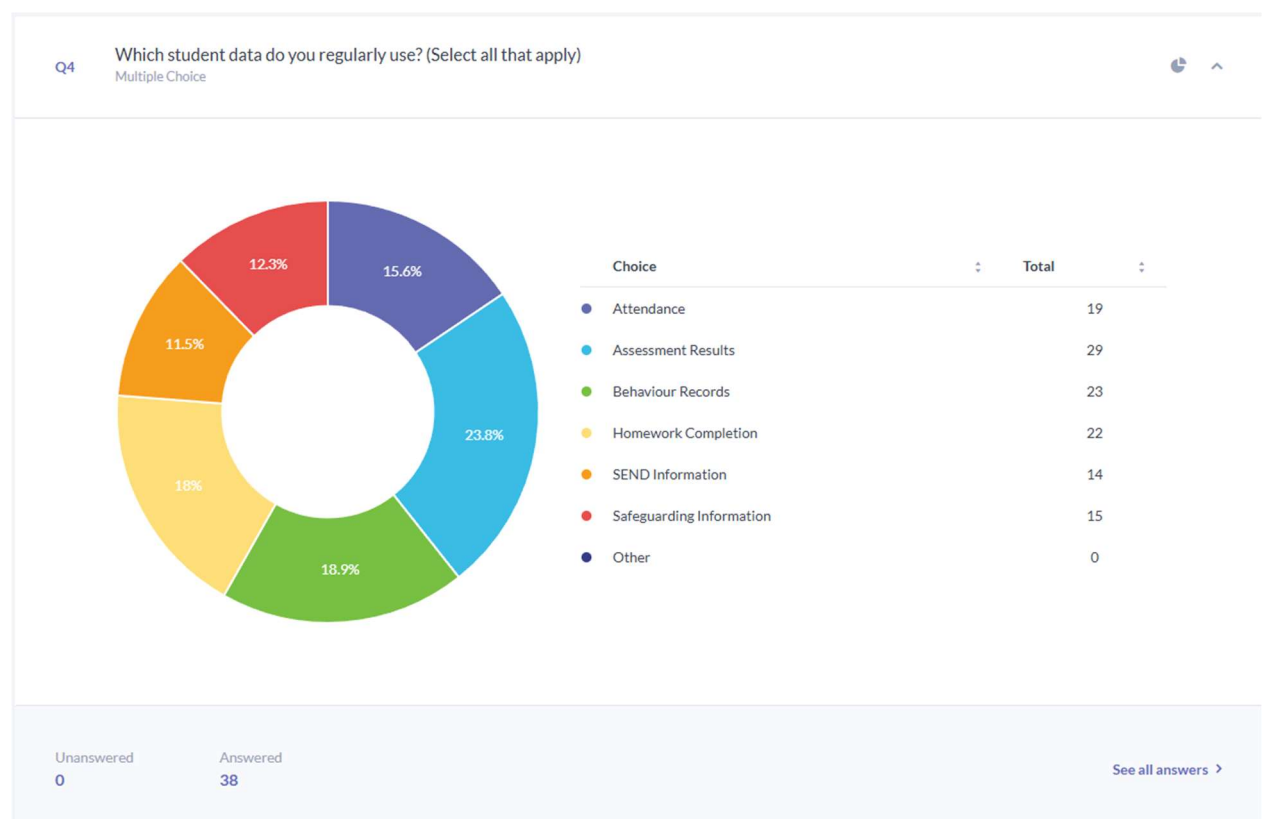
**What is the most frustrating part of using student data in your current role?**

- ☐ Free-text response
- ☐ **If you could add one feature to existing systems such as SIMS, Arbor or Bromcom, what would it be?**
- ☐ Free-text response

## Appendix B.2 Survey Respondent Summary

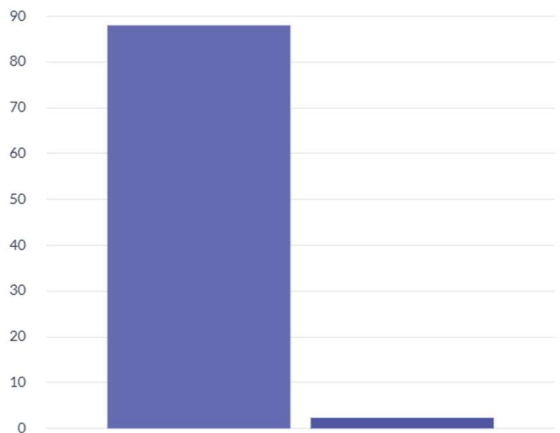
| Characteristic      | Responses |
|---------------------|-----------|
| Total responses     | 38        |
| Classroom Teachers  | 24        |
| Heads of Department | 7         |
| Senior Leaders      | 2         |
| Teaching Assistants | 4         |
| Other               | 1         |
| Primary             | 10        |
| Secondary           | 28        |

## Appendix B.3 Selected Survey results



Q7 How easy is it to identify students who need support using your current systems?

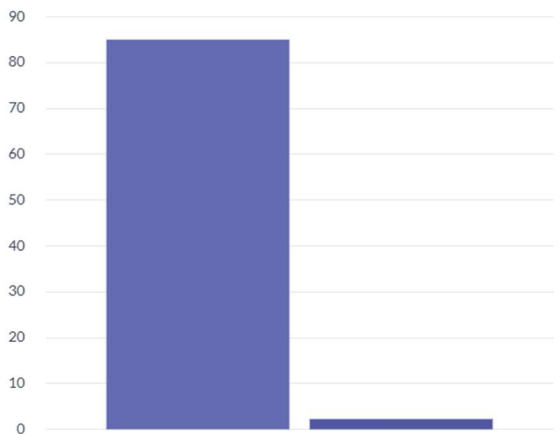
Scoring



| Choice | Score | Average |
|--------|-------|---------|
| Score  | 88    | 2.32    |

Q9 "School data is collected but does not always lead to meaningful action."

Scoring



| Choice | Score | Average |
|--------|-------|---------|
| .      | 85    | 2.24    |

Unanswered  
0

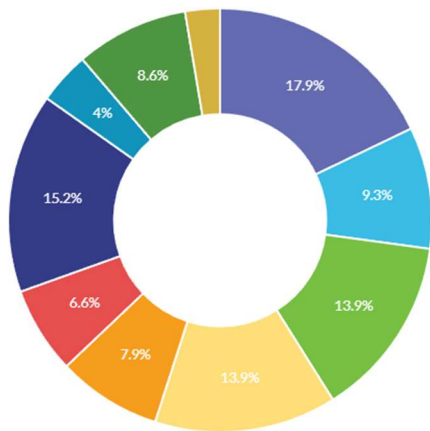
Answered  
38

[See all answers >](#)

Q11

Which features would be most useful in a student analytics dashboard? (Select up to 5)

Multiple Choice

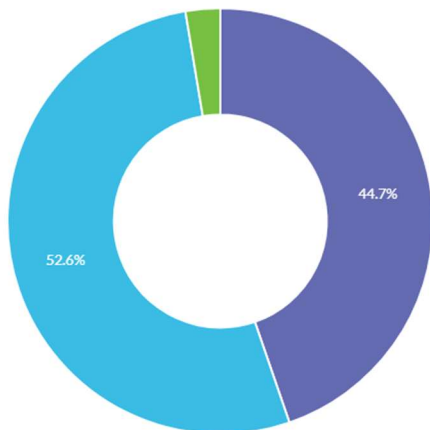


| Choice                                 | Total |
|----------------------------------------|-------|
| At-risk student alerts                 | 27    |
| Attendance trend analysis              | 14    |
| Behaviour trend analysis               | 21    |
| Performance predictions                | 21    |
| Intervention recommendations           | 12    |
| Parent email generation                | 10    |
| Student summaries                      | 23    |
| Automated reports                      | 6     |
| AI explanations of performance changes | 13    |
| Intervention tracking                  | 4     |

Q13

Would you trust an AI recommendation if the system clearly explained how it reached the recommendation?

Multiple Choice



| Choice | Total |
|--------|-------|
| Yes    | 17    |
| Maybe  | 20    |
| No     | 1     |

| Date       | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 days ago | Ensuring the behaviour is good as well as performing well in class with no interruption.                                                                                                                                                                                                                                                                                                                                                   |
| Jun 18     | There are times that assessment results are not reflective of a student's ability, engagement in lesson and their capability. Rather, the student lacks confidence to perform in exam conditions. These students are considered for set change because of the poor results. Decisions then need to be made on what is best for the particular student. Interventions can be put in place to support them or class change may be necessary. |
| Jun 18     | Turning data into clear, actionable steps for supporting individual student needs.                                                                                                                                                                                                                                                                                                                                                         |
| Jun 18     | Identifying meaningful trends in the data and translating those insights into effective instructional strategies for different student needs.                                                                                                                                                                                                                                                                                              |
| Jun 17     | Ensuring access is easy, straightforward and simple to understand.                                                                                                                                                                                                                                                                                                                                                                         |
| Jun 17     | Classrooms often contain students with varying abilities and learning needs, making it challenging to use data to differentiate instruction effectively for everyone.                                                                                                                                                                                                                                                                      |
| Jun 17     | Ensuring student data is accurate, complete, and up to date.                                                                                                                                                                                                                                                                                                                                                                               |
| Jun 16     | If through positive discrimination pupils of lower ability have been placed in higher sets, it can be difficult to give them the constant support they need as the content cannot be diluted for a higher ability class                                                                                                                                                                                                                    |
| Jun 16     | The data is oversimplified as it is colour coded- for example students of concern are highlighted in red. It would be helpful if you were easily able to see how much students are underperforming by                                                                                                                                                                                                                                      |
| Jun 16     | .                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Jun 16     | Mixed ability students                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Jun 16     | Grasping the correct methods to aid their learning according to their needs and help to the extent that they need help.                                                                                                                                                                                                                                                                                                                    |
| Jun 16     | This year based on data we made a student sit a foundation paper but I think she could've sit the higher paper. Sometimes the data doesn't take into account their effort they put in and potential improvement they can make. I really do think we made a mistake with her and limited their results                                                                                                                                      |
| Jun 16     | Interpreting data                                                                                                                                                                                                                                                                                                                                                                                                                          |